

AV Fundamentals for Educators

Instructor Guide

ISTE x AVIXA

A Co-Branded Curriculum Resource for Classroom Teachers,
Instructional Designers, and Course Facilitators

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Prepared for: Community Colleges • University Professional Development Programs
K-12 Technology Coordinators • Corporate Training Facilitators

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Front Matter

How to Use This Instructor Guide

This Instructor Guide is the primary resource for educators facilitating *AV Fundamentals for Educators*, a co-branded curriculum produced by ISTE (International Society for Technology in Education) and AVIXA (Audiovisual and Integrated Experience Association). It is designed to be read before the course begins and referenced week by week during instruction. The guide does not replace the student textbook — it supplements it with pedagogical scaffolding, facilitation notes, assessment instruments, and differentiation strategies.

The guide is organized into five functional zones:

1. **Front Matter** — Philosophy, configurations, prerequisites, and pre-course checklists
2. **Master Pacing Guide** — Week-by-week schedule across all delivery formats
3. **Chapter Instructor Notes** — Lesson plans, discussion questions, labs, misconceptions, rubrics
4. **Assessment Instruments** — Midterm, final, and rubric masters ready for reproduction
5. **Supplemental Resources** — Guest speakers, PD, certification pathway, SME contacts



Instructor Tip

Read the full guide before Week 1. Pay particular attention to the Differentiation Strategies in each chapter — many cohorts include a mix of complete beginners and AV professionals, and this range is the most common facilitation challenge instructors report.

All assessment instruments are reproducible for classroom use under the ISTE x AVIXA co-branded license. Answer keys are marked **INSTRUCTOR ONLY** and should not be distributed to students. Digital versions of all rubrics are available in editable format through the ISTE LMS resource portal.

Course Philosophy

Principles-First

This course is deliberately structured around enduring principles rather than specific products. The AV industry evolves rapidly — display technologies, signal formats, and control protocols that are current today may be obsolete in five years. By grounding instruction in underlying physical and logical principles (acoustics, optics, signal flow, network topology), students develop transferable knowledge that remains valid across technological generations. Instructors should reinforce this stance explicitly: when a student asks "Which brand should we buy?" redirect to "What performance criteria matter for this context, and how do we evaluate any product against them?"

Equity-Centered

AV technology in educational settings profoundly affects student access to learning. Poor room acoustics, inadequate captioning, inaccessible control interfaces, and inequitably distributed technology infrastructure all function as barriers. This course foregrounds equity at every chapter — not as a standalone unit but as a lens applied to every design decision. Instructors are expected to model this stance and to challenge assumptions about whose needs are treated as the default.

Vendor-Agnostic

No manufacturer, integrator, or platform receives preferential treatment in this curriculum. Where product examples are necessary for illustration, multiple competing options are presented. Instructors who have personal relationships with vendors are encouraged to maintain that neutrality in the classroom. Guest speakers from industry are welcome but should be briefed on the vendor-agnostic policy before presenting.

Recommended Course Configurations

Format	Duration	Session Structure	Best For	Notes
14-Week Semester	14 weeks	2 × 75-min sessions/week or 3 × 50-min sessions/week	Community college, university	Full chapter coverage; recommended default configuration
12-Week Semester	12 weeks	2 × 75-min sessions/week	Trimester programs, accelerated semesters	Combine Chapters 2&3 (Weeks 3) and Chapters 7&8 (Week 9); condense Weeks 13&14
8-Week Intensive	8 weeks	2 chapters/week ; 3 × 75-min sessions/week	Boot camps, summer institutes, corporate cohorts	Reduce lab time; prioritize chapters 1, 2, 5, 7, 11 for minimum viable coverage
Self-Paced	Flexible (8-20 weeks)	Asynchronous with scheduled synchronous check-ins	Online PD, independent study, K-12 coordinator cohorts	Convert all labs to virtual simulations or documented walkthroughs; use discussion boards for exit tickets

Important — 12-Week Adaptation

When compressing to 12 weeks, do not eliminate the Accessibility chapter (Ch. 11). Compress Chapter 12 (Emerging Tech) to a single session discussion instead. Accessibility is core to ISTE alignment and should never be treated as optional content.

Required Materials and Prerequisites

Student Materials

- *AV Fundamentals for Educators* textbook (ISTE × AVIXA, Edition 1.0) — required
- Lab Workbook (included with textbook or available as PDF download from ISTE portal)
- Scientific calculator or equivalent smartphone app
- Access to a device capable of running a modern web browser (for virtual labs and AVIXA Learning Portal)
- Notebook or digital note-taking tool for reflection journals

Instructor Materials

- This Instructor Guide (Edition 1.0)
- Instructor Answer Key Supplement (separate document — request from ISTE/AVIXA)
- AVIXA Learning Management System (LMS) instructor access
- ISTE Educator Standards reference card (available at iste.org)
- Access to a functional AV demonstration system for labs (see Technology Requirements below)

Recommended Student Prerequisites

- Basic computer literacy (file management, web navigation, document creation)
- Familiarity with classroom or meeting room AV as an end user
- No prior technical AV knowledge required — course is designed for beginners with scaffolding for experienced learners

Technology Requirements for Instruction

Component	Minimum Requirement	Recommended	Purpose
Display System	Single projector or flat panel, 1080p	Dual displays (1 for instruction, 1 for demo)	Slides, signal flow demos
Audio System	Amplified speakers, USB microphone	Installed speaker array + wireless handheld mic	Audio labs, SPL demonstrations
Signal Cabling Kit	HDMI, USB-C, VGA, 3.5mm, XLR samples	Full connector sample kit (30+ types)	Hands-on connector identification labs
Network Access	Wi-Fi, 25 Mbps per student minimum	Wired Ethernet stations + Wi-Fi	Networking labs, AVIXA LMS
Control System Demo	Software-based control panel (tablet app)	Physical touch panel + control processor	Ch. 6 control systems lab
Video Conferencing	Laptop + built-in webcam + browser-based VC	Dedicated room system (Teams/Zoom Room)	Ch. 7 collaboration lab
Digital Signage Player	Software player on desktop PC	Dedicated media player hardware + secondary display	Ch. 9 digital signage lab
Measuring Tools	SPL meter app (smartphone), tape measure	Handheld SPL meter, laser distance measure	Acoustic and room measurement labs

Accessibility Checklist for Instructors (Pre-Course Setup)

Note

Complete this checklist before the first class session. Revisit it at the midpoint of the course. Many of these items must be coordinated with your institution's facilities or IT department weeks in advance.

#	Item	Action Required	Completed
1	Captioning availability for all video content	Verify captions on all video resources; request CART if needed	☐
2	Screen reader compatibility for LMS materials	Test all uploaded PDFs and slides with NVDA or JAWS	☐
3	Physical lab stations accessible to wheelchair users	Confirm table height and aisle clearance (min. 36 in.)	☐
4	Color-blind-safe slide designs	Run slides through a color contrast checker (WCAG AA minimum)	☐
5	Audio amplification for instructor voice	Use handheld or lapel mic for all instruction	☐
6	Alternative formats for textbook	Coordinate with disability services for large-print or digital accessible editions	☐
7	Extended time accommodations for assessments	Confirm protocol with your institution and note affected students	☐

#	Item	Action Required	Completed
8	Lab partner pairing for students with motor limitations	Plan flexible groupings that do not isolate students with disabilities	☐
9	Lighting adequacy for lip-reading and signing	Confirm no backlighting during instruction; supplement if needed	☐
10	Emergency evacuation route communicated	Review with students on Day 1; include lab-specific egress	☐

Master Pacing Guide — 14-Week Course Schedule

The table below is the authoritative course calendar for the 14-week configuration. Weeks 12 and 14 may be compressed for the 12-week format as noted in the Course Configurations section. ISTE Standard codes refer to the ISTE Standards for Educators (2024 v4.02). AVIXA Domain codes refer to the CTS Exam Content Outline domains.

Wk	Chapter / Topic	Key Learning Objectives	Required Reading	Lab Activity	Assessment	ISTE Standard	AVIXA Domain
1	Course Introduction; Chapter 1 Intro: The AV Ecosystem	Identify stakeholders in educational AV; define AV as a discipline;	Ch. 1, pp. 1–18; ISTE Educator Standards Overview	Lab 1A: AV System Scavenger Hunt (campus walk)	Syllabus quiz; Intro survey	2.1 Learner; 2.4 Collaborator	Domain 1: AV Industry

Wk	Chapter / Topic	Key Learning Objectives	Required Reading	Lab Activity	Assessment	ISTE Standard	AVIXA Domain
		articulate course outcomes					
2	Chapter 1 continued: AV Ecosystem (roles, standards, industry bodies)	Distinguish AVIXA, ISTE, ISO, and IEC roles; map AV career pathways; explain standards vs. codes	Ch. 1, pp. 19–36; AVIXA Industry Outlook (excerpt)	Lab 1B: AV Career Pathway Mapping	Chapter 1 Quiz	2.1 Learner; 2.3 Citizen	Domain 1: AV Industry
3	Chapter 2: Audio Fundamentals	Explain acoustic principles; calculate SPL; identify transducer types; describe signal chain	Ch. 2, pp. 37–68	Lab 2: SPL Measurement and Room Acoustics	Chapter 2 Quiz	2.5 Designer	Domain 2: Audio
4	Chapter 3: Video Fundamentals	Explain image formation; calculate aspect ratios and throw distances; compare	Ch. 3, pp. 69–104	Lab 3: Projector Throw and Luminance Testing	Chapter 3 Quiz	2.5 Designer	Domain 3: Video

Wk	Chapter / Topic	Key Learning Objectives	Required Reading	Lab Activity	Assessment	ISTE Standard	AVIXA Domain
		display technologies					
5	Chapter 4: Signal Flow	Trace signal paths; identify connectors and cable types; troubleshoot basic signal faults	Ch. 4, pp. 105–136	Lab 4: Signal Flow Diagram and Cable Termination	Chapter 4 Quiz	2.5 Designer; 2.6 Facilitator	Domain 4: Signal Distribution
6	Chapter 5: Networking for AV	Apply IP addressing basics; differentiate AV-over-IP protocols; explain network security basics for AV	Ch. 5, pp. 137–170	Lab 5: IP Address Configuration and Network Map	Chapter 5 Quiz	2.5 Designer	Domain 5: Networking
7	Chapter 6: Control Systems + Midterm Review	Describe control system topologies; program a basic macro; conduct midterm review across Chs. 1–6	Ch. 6, pp. 171–202; Midterm Study Guide	Lab 6: Control Panel Macro Programming	Chapter 6 Quiz; Midterm Review Game	2.5 Designer; 2.6 Facilitator	Domain 6: Control Systems

Wk	Chapter / Topic	Key Learning Objectives	Required Reading	Lab Activity	Assessment	ISTE Standard	AVIXA Domain
8	Midterm Exam Week	Demonstrate mastery of Chapters 1-6 content and skills	Review all Chs. 1-6	No new lab — open lab review session available	Midterm Exam (40 items)	All 2.1-2.7	Domains 1-6
9	Chapter 7: Collaboration Technology	Configure a UC&C endpoint; compare BYOD vs. room systems; apply hybrid equity principles	Ch. 7, pp. 203-238	Lab 7: Hybrid Meeting Room Setup and Test	Chapter 7 Quiz	2.4 Collaborator; 2.6 Facilitator	Domain 7: Collaboration
10	Chapter 8: Classroom AV Design	Apply ANSI/INFOCOMM standards for sight lines and audio coverage; produce a basic room design	Ch. 8, pp. 239-276	Lab 8: Classroom AV Design Exercise	Chapter 8 Quiz	2.5 Designer; 2.7 Analyst	Domain 8: AV Design
11	Chapter 9: Digital Signage	Design a compliant digital signage layout;	Ch. 9, pp. 277-308	Lab 9: Digital Signage Content Creation and Scheduling	Chapter 9 Quiz	2.5 Designer; 2.3 Citizen	Domain 9: Digital Signage

Wk	Chapter / Topic	Key Learning Objectives	Required Reading	Lab Activity	Assessment	ISTE Standard	AVIXA Domain
		explain CMS architecture; evaluate content for accessibility		ng			
12	Chapter 10: Maintenance and Troubleshooting	Execute a preventive maintenance routine; apply systematic troubleshooting methodology; document service records	Ch. 10, pp. 309-342	Lab 10: Preventive Maintenance Checklist and Fault Simulation	Chapter 10 Quiz	2.7 Analyst	Domain 10: Operations
13	Chapter 11: Accessibility in AV	Apply ADA and WCAG 2.1 to AV systems; evaluate a room for accessibility gaps; propose remediation	Ch. 11, pp. 343-376; ADA Title II guidance (excerpt)	Lab 11: Accessibility Audit of a Campus AV Space	Chapter 11 Quiz	2.3 Citizen; 2.5 Designer	Domain 11: Accessibility
14	Chapter 12:	Evaluate AI, XR,	Ch. 12, pp. 377-	Capstone	Final Exam	2.1 Learner;	Domains 1-11

Wk	Chapter / Topic	Key Learning Objectives	Required Reading	Lab Activity	Assessment	ISTE Standard	AVIXA Domain
	Emerging Technologies + Final Exam Prep + Capstone Presentations	and IoT applications in educational AV; synthesize course learning in capstone; complete final exam	410; Final Exam Study Guide	Presentations (Lab 12)	(50 items); Capstone Project Due	2.7 Analyst	(cumulative)

Chapter-by-Chapter Instructor Notes

Chapter 1: The AV Ecosystem

1.1 Instructor Overview

Chapter 1 establishes the conceptual framework for the entire course. Instructors should resist the temptation to treat this chapter as a glossary recitation. The goal is for students to construct a mental map of how AV technology, educational practice, professional organizations, and standards bodies all interrelate. Spend significant time on the distinction between AV as infrastructure (the room systems and cabling) and AV as pedagogy (how those systems enable or constrain learning). A common instructor pitfall is moving too quickly through the AVIXA and ISTE organizational landscapes — these bodies are foundational to the course's dual-

credential structure, and students need to understand why both matter. Use the career pathway mapping lab to anchor abstract concepts in real occupational outcomes. Emphasize that this is a principles-first, vendor-agnostic course so students know what to expect from Week 1. Address any student anxiety about technical prerequisites directly: this course begins with no technical assumptions.

1.2 Lesson Plans

Session A — AV in Education: Why It Matters

Time	Activity	Details
0–5 min	Warm-Up	Show 2 images: an under-resourced classroom with broken equipment and a well-designed hybrid learning space. Ask: "What do you notice? What's different about the learning experience in each?"
5–25 min	Direct Instruction	Overview of AV ecosystem: definition of AV, the role of AVIXA and ISTE, career pathways (integrator, designer, educator, coordinator), standards bodies (ANSI, IEC, ISO). Use a concept map on the board.
25–40 min	Active Learning	Lab 1A: Campus AV Scavenger Hunt. Students walk in pairs to 5 campus locations, photographing and cataloging AV systems. Return and share findings.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Name one AV system you saw today and identify one thing you don't yet understand about how it works."

Session B — Standards, Industry Bodies, and Professional Roles

Time	Activity	Details
0-5 min	Warm-Up	Quick poll: "Have you ever experienced an AV failure during a class or meeting? What happened?" Share 2-3 responses.
5-25 min	Direct Instruction	AVIXA certification pathways (AV Technologist → CTS → CTS-D / CTS-I); ISTE Educator Standards overview (2024, v4.02); role of standards (ANSI/INFOCOMM 1M, ANSI/AVIXA) in AV design. Explain how this course maps to both frameworks.
25-40 min	Active Learning	Lab 1B: Career Pathway Mapping. Students use provided template to map their current role → desired AV competency → applicable credential. Pair-share results.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Which ISTE Educator Standard feels most relevant to your current work, and why?"

1.3 Discussion Questions

6. How does the presence or absence of reliable AV infrastructure create or reinforce educational inequity? Provide a specific example from your own experience or observation.
7. In what ways does the vendor-agnostic philosophy of this course align with or conflict with how AV purchasing decisions are actually made at your institution?

8. Compare and contrast the roles of AVIXA and ISTE. How does each organization approach the relationship between technology and education differently?
9. What professional standards — in your current field or in AV — exist to protect end users? Who enforces them, and what happens when they are not followed?
10. The AV industry is increasingly converging with IT networking. What does this mean for educators who manage classroom technology? What new competencies does it require?

1.4 Differentiation Strategies

Learner Profile	Strategy
Beginners (no AV background)	Provide a glossary card for Week 1 terms. Use the scavenger hunt to build observational vocabulary before introducing technical terms. Pair with a more experienced peer for Lab 1B.
Experienced teachers (limited AV knowledge)	Connect AV ecosystem to familiar pedagogical frameworks (UDL, backwards design). Ask them to assess their current classroom AV using the accessibility checklist.
AV specialists (industry background)	Challenge them to identify where their industry knowledge has gaps in educational context. Assign supplemental reading on ISTE standards. Ask them to serve as a resource for peers during lab — but not to dominate.

1.5 Lab Facilitation Notes

Lab 1A Setup: Identify 5 campus locations with diverse AV systems (lecture hall, conference room, huddle space, corridor display, outdoor PA) before class. Create a simple observation sheet with fields: Location, System Type, Estimated Age, Functionality Status, Accessibility Features Observed. Brief security or facilities staff if students will be entering restricted areas.

Lab 1A Run: Assign pairs, distribute observation sheets, set a 15-minute return time. Keep one location on campus with a known fault for discussion value.

Lab 1A Debrief: Gallery walk of completed sheets. Ask: Which system appeared most equitably designed? Which appeared most problematic? Why?

1.6 Common Student Misconceptions

Misconception	Correction Strategy
"AV is just IT with screens and speakers."	Explain the distinct discipline of AV design — acoustics, human factors, signal integrity, and room geometry are not addressed by standard IT curricula. Show the separate AVIXA and CompTIA credential tracks.
"Better/more expensive equipment always means better AV."	Use examples of high-budget classrooms with poor acoustic treatment failing worse than modest systems in well-designed rooms. Introduce the principle that design precedes hardware selection.
"AVIXA certification is only for integrators, not educators."	Share case studies of K-12 technology coordinators and instructional designers who hold CTS credentials and how it improved their procurement and design conversations.
"AV standards are optional guidelines."	Distinguish between voluntary standards (ANSI/AVIXA) and code-enforceable requirements (ADA, local building codes). Explain the legal and liability implications.
"This course is mainly about clicking buttons on projectors."	Redirect early and often: this course is about principles that allow you to evaluate, design, and advocate for AV systems — not just operate them. Reference the capstone project.

1.7 Assessment Alignment Table

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q1	Definition of	Domain 1	2.1	Remember

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
	AV			
Q2	AVIXA mission and membership	Domain 1	2.1	Understand
Q3	ISTE Educator Standards overview	Domain 1	2.1, 2.3	Understand
Q4	AV vs. IT distinctions	Domain 1	2.5	Analyze
Q5	Career pathway identification	Domain 1	2.1	Apply
Q6	Standards bodies (ANSI, IEC, ISO)	Domain 1	2.3	Remember
Q7	Voluntary vs. mandatory standards	Domain 1	2.3	Analyze
Q8	Equity in AV infrastructure	Domain 1	2.3	Evaluate
Q9	Vendor-agnostic evaluation	Domain 1	2.7	Evaluate
Q10	AVIXA AV Technologist pathway	Domain 1	2.1	Apply

1.8 Accommodation Notes

The campus scavenger hunt involves physical mobility. Provide a seated alternative: a virtual tour using pre-captured images or a video walkthrough of each AV space. Students with low vision should be paired with a sighted partner for physical navigation; ensure observation sheets are available in large-print format. For the career mapping lab, offer a verbal recording option in lieu of written documentation for students with writing-related disabilities.

1.9 Chapter Lab Rubric — Lab 1: AV Ecosystem Exploration

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All system identifications are correct; terminology used precisely	Most identifications correct; minor terminology errors	Several misidentifications; terminology inconsistent	Significant errors; terminology absent or incorrect throughout
Procedure	All 5 locations documented; observation sheet complete and organized	4 of 5 locations documented; minor omissions	3 locations documented; several fields incomplete	Fewer than 3 locations; sheet largely incomplete
Documentation	Observations written in clear, specific language; accessibility notes thorough	Observations generally clear; some vague descriptions	Observations sparse or overly general	Minimal documentation; observations not usable
Teamwork	Roles clearly divided; both partners contributed equally; respectful collaboration evident	Mostly equitable contribution; one partner slightly dominant	One partner did most of the work; limited collaboration	No evidence of collaboration; individual work submitted as pair work

Chapter 2: Audio Fundamentals

2.1 Instructor Overview

Chapter 2 is the first deeply technical chapter and the point where students with no science background often feel the most anxiety. Instructors should begin by normalizing this discomfort and making the physics tangible: sound is pressure — you can feel it. Use physical demonstrations aggressively. Clap your hands, use a tone generator on your phone, walk students through the inverse square law with a ruler and a sound source. The most important conceptual milestone for this chapter is the decibel: students frequently confuse it as a unit of volume rather than a

logarithmic ratio. Spend a full segment on this misconception before moving to transducers. The signal chain (source → amplification → transduction → transmission → destination) is the second major framework — revisit it in every subsequent chapter to build cumulative understanding. Do not rush through room acoustics; this is the concept most directly applicable to every educator's daily professional life and is the richest ground for equity conversations about under-resourced classrooms with poor acoustic environments.

2.2 Lesson Plans

Session A — Sound Physics and the Decibel

Time	Activity	Details
0–5 min	Warm-Up	Play two audio clips at different SPLs. Ask: "How much louder is the second clip? Twice as loud? Ten times?" Introduce the challenge of measuring loudness.
5–25 min	Direct Instruction	Acoustic fundamentals: frequency, wavelength, amplitude, phase. The decibel as a logarithmic ratio (dB SPL, dBu, dBV). Inverse square law. Human hearing range and thresholds. Fletcher-Munson curves (brief introduction).
25–40 min	Active Learning	Decibel calculation worksheet. Students calculate SPL at various distances using the inverse square law. Check answers in pairs before instructor review.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Why is the decibel scale logarithmic rather than linear? Give one real-world example of why this matters."

Session B — Transducers, Signal Chain, and Room Acoustics

Time	Activity	Details
0-5 min	Warm-Up	Pass around a microphone capsule and a speaker driver (removed from housing if possible). Ask: "What do these two things have in common? What's different?"
5-25 min	Direct Instruction	Transducer types: dynamic, condenser, ribbon microphones; woofer, tweeter, full-range speakers. Audio signal chain. Balanced vs. unbalanced signals. Introduction to room acoustics: reflection, absorption, diffusion, RT60.
25-40 min	Active Learning	Lab 2 (Part 1): SPL measurement exercise using smartphone SPL meter app. Students measure ambient noise and instructor speech level at 3 distances in the classroom. Record and compare to ANSI classroom standards.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Your classroom measures 58 dB ambient noise. The ANSI standard for classroom background noise is 35 dB (NC-35). What are two possible sources of the excess noise, and what might be done?"

2.3 Discussion Questions

11. ANSI/ASA S12.60 sets maximum background noise levels for classrooms at NC-35. Research suggests many public school classrooms exceed NC-50. What are the academic impact implications, particularly for students with hearing loss or auditory processing differences?
12. A school district is choosing between a distributed speaker system and a single centrally-mounted speaker for a cafeteria/commons area. What acoustic factors should guide the decision, and how does the use case affect the answer?
13. How does the inverse square law apply to equity of experience in a large lecture hall? Who benefits most from the physics — students in the front or the back? What design strategies can compensate?
14. A teacher routinely uses their laptop's built-in speakers for audio playback in a 30-student classroom. Using what you know about SPL and coverage, what problems can you predict, and what solutions would you recommend?
15. Vocal fatigue is a documented occupational hazard for teachers. How does understanding the audio signal chain inform the case for installing classroom sound field amplification systems?

2.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Provide a decibel reference card (whisper=30dB, conversation=60dB, concert=110dB). Use visual analogies: frequency = color, amplitude = brightness. Allow calculators for all math exercises.
Experienced teachers	Connect acoustic concepts directly to classroom management: "When you raise your voice to reach the back row, you are fighting the inverse square law." This framing makes physics immediately practical.
AV specialists	Assign supplemental reading on STIPA

Learner Profile	Strategy
	(Speech Transmission Index for PA systems) and ask them to evaluate whether the campus PA system meets intelligibility standards. Have them present findings to the class.

2.5 Lab Facilitation Notes

Setup: Download a calibrated SPL meter app (NIOSH SLM or Decibel X recommended) on instructor's device and test before class. Identify 3 measurement positions in the room: front, middle, rear. Prepare a tone generator to play a 1 kHz tone at a consistent level for controlled measurements. Have printed copies of ANSI/ASA S12.60 classroom noise criteria available for reference.

Run: Demonstrate proper SPL meter technique (hold at ear height, away from body, point toward source). Students rotate through measurement positions in groups of three. Record all values on a shared class chart (whiteboard or projected spreadsheet).

Debrief: Compare class measurements to standard. Discuss sources of discrepancy. Introduce the concept that designing a classroom AV system without knowing the ambient noise floor is designing blind.

2.6 Common Student Misconceptions

Misconception	Correction Strategy
"More volume always improves speech intelligibility."	Demonstrate with a recording played at high volume in a reverberant space — louder makes it worse. Introduce the concept of signal-to-noise ratio and reverb time.
"Decibels are a unit of volume — 20 dB is twice as loud as 10 dB."	Use the logarithmic scale explicitly: 10 dB increase = perceived doubling of loudness; 6 dB = halving of distance. Have students calculate 3-step examples.
"Foam panels on walls will fix any acoustic problem."	Explain frequency-dependent absorption: thin foam panels absorb high frequencies but have minimal effect on low-frequency rumble and HVAC noise. Proper acoustic treatment

Misconception	Correction Strategy
	requires analysis first.
"A wireless microphone is always better than a wired one."	Discuss RF congestion in education environments, battery management, and latency. Explain that wireless adds complexity and failure modes — choice depends on use case.
"Balanced cables eliminate all noise."	Explain common-mode rejection: balanced reduces induced noise but does not eliminate all interference. Proper grounding, cable routing away from power, and appropriate cable length remain important.

2.7 Assessment Alignment Table

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q1	Definition of frequency and wavelength	Domain 2	2.5	Remember
Q2	Decibel as logarithmic ratio	Domain 2	2.5	Understand
Q3	Inverse square law calculation	Domain 2	2.5	Apply
Q4	Transducer types and function	Domain 2	2.5	Understand
Q5	Balanced vs. unbalanced signals	Domain 2, 4	2.5	Analyze
Q6	Room acoustics: reflection, absorption	Domain 2	2.5	Understand
Q7	RT60 definition and significance	Domain 2	2.5	Understand
Q8	ANSI classroom noise standard	Domain 2	2.3	Apply

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q9	Sound field amplification in classrooms	Domain 2	2.5, 2.3	Evaluate
Q10	Speaker coverage pattern selection	Domain 2	2.5	Analyze

2.8 Accommodation Notes

Students who are Deaf or hard of hearing may have limited ability to perceive SPL variation directly; ensure they can complete the lab using a visual SPL meter display rather than auditory perception. Provide tactile demonstrations where appropriate (touching a speaker cone to feel vibration). For students with sound sensitivity, provide advance notice of tone generator use and offer earplugs or allow observation from a greater distance. Lab materials should include captioned video demonstrations of all audio concepts.

2.9 Chapter Lab Rubric — Lab 2: SPL Measurement and Room Acoustics

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All SPL measurements within ± 2 dB of instructor baseline; calculations correct	Measurements within ± 4 dB; minor calculation errors	Measurements vary significantly; conceptual errors in calculations	Measurements unreliable; calculations largely incorrect
Procedure	All 3 measurement positions completed; correct meter technique observed	All positions attempted; one technique deviation noted	2 of 3 positions completed; technique inconsistent	Incomplete measurement set; significant technique errors
Documentation	Data table complete; comparison to	Data table complete; brief	Data table partially complete; no	Minimal documentation; data not

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	standard with written analysis included	comparison present	comparison to standard	usable
Teamwork	Roles rotated among all group members; all contributed to analysis	Roles mostly shared; one member slightly less engaged	One or two members dominated; others disengaged	No evidence of shared participation

Chapter 3: Video Fundamentals

3.1 Instructor Overview

Chapter 3 mirrors the structure of Chapter 2 — moving from physics to transducers to systems — but in the visual domain. Students typically find video concepts more intuitive because they interact with displays daily; the challenge is moving them from consumer-level understanding to professional evaluation criteria. Key teaching points: the difference between resolution (pixel count) and perceived image quality (which also depends on viewing distance, luminance, and contrast ratio); the concept of throw ratio as the basis for projector placement; and the distinction between display technologies (LCD, LED, laser) at the principle level rather than brand level. A very common misconception is that "4K is always better" — use this as an entry point into the viewing distance formula and the concept of visual acuity limits. The color space and HDR segments are often more difficult than instructors expect — spend extra time and use side-by-side visual comparisons. End the chapter with the classroom application: sight line analysis, which connects directly to Chapter 8.

3.2 Lesson Plans

Session A — Light, Resolution, and Display Technologies

Time	Activity	Details
0-5 min	Warm-Up	Display two images side-by-side: 1080p viewed at 6 feet vs. 4K viewed at 30 feet. Ask: "Which looks better from where you're sitting?"
5-25 min	Direct Instruction	Light fundamentals: wavelength and color perception. Pixel, resolution, and aspect ratio. Common resolutions (1080p, 4K, 8K). Display technologies: LCD, LED, OLED, laser. Luminance (nits/lumens) and contrast ratio. Viewing distance formula.
25-40 min	Active Learning	Calculate optimal viewing distances for 3 display scenarios using the 1.5H and 6H rule. Compare calculations to actual room measurements. Discuss: is the display in this room appropriately sized?
40-50 min	Debrief / Exit Ticket	Exit ticket: "A classroom is 40 feet deep. What is the minimum screen height recommended, and what resolution would be discernible at that distance?"

Session B — Projectors, Cameras, and Color

Time	Activity	Details
0-5 min	Warm-Up	Shine a projector on the board with the room lights fully on, then off. Ask: "What changed? What does this tell you about lumens and ambient light?"
5-25 min	Direct Instruction	Projector throw ratios

Time	Activity	Details
		(standard, short-throw, ultra-short-throw). Lumens vs. ANSI lumens. Camera fundamentals: sensor size, frame rate, codec basics. Color spaces (sRGB, DCI-P3). Introduction to HDR. Calibration concepts.
25–40 min	Active Learning	Lab 3: Projector Throw Calculation. Students use the throw ratio formula to determine projector placement for 3 hypothetical room configurations. Check with online throw ratio calculators.
40–50 min	Debrief / Exit Ticket	Exit ticket: "A projector has a throw ratio of 1.5:1. The desired image width is 120 inches. How far from the screen must the projector be placed?"

3.3 Discussion Questions

16. A school board is debating replacing all classroom projectors with flat-panel displays. What criteria should guide that decision beyond purchase price? Who should be involved in the evaluation?
17. How does ambient light affect the equity of the learning experience in a classroom with windows on one side? What design strategies can mitigate this?
18. Ultra-short-throw projectors have become popular in K–12 settings. What are the trade-offs compared to standard-throw projectors from an image quality, maintenance, and total cost of ownership perspective?
19. Video conferencing cameras in hybrid classrooms typically capture the "front of room" perspective. How does this affect remote students' equity of experience, and what camera approaches can address it?

20. Color calibration in educational AV is rarely performed after initial installation.

What are the implications for color-accuracy-dependent disciplines (art, science, medical education)?

3.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Start with consumer language ("TV screen size", "streaming resolution") and build to technical terms. Provide a one-page throw ratio formula card. Allow use of online calculators after demonstrating manual calculation.
Experienced teachers	Ask them to sketch their classroom and identify whether the existing display meets viewing distance criteria. This creates immediate personal relevance and produces data useful for the capstone project.
AV specialists	Assign advanced reading on HDR standards (HDR10, Dolby Vision) and color volume. Ask them to explain the difference between peak luminance and sustained luminance to the class in Session B.

3.5 Lab Facilitation Notes

Setup: Pre-load a throw ratio spreadsheet template on classroom computers. Prepare 3 room configuration cards (room dimensions, intended image size, available mounting positions). Have at least two different projectors available if possible — one standard-throw and one short-throw — for physical comparison.

Run: Walk through the throw ratio formula together before students work independently. Emphasize units consistency (all in inches or all in meters). If a second projector is available, have students measure actual throw distance and compare to calculated value.

Debrief: Discuss cases where the calculation result is physically impossible given room constraints. What are the solutions? (Different lens, different mount position, different image size, different technology.)

3.6 Common Student Misconceptions

Misconception	Correction Strategy
"Higher resolution always means better image quality."	Demonstrate the viewing distance formula. At typical classroom viewing distances, the human eye cannot distinguish 4K from 1080p beyond approximately 12 feet from the screen for a standard screen size.
"Lumens of a projector = brightness of the image."	Explain that image brightness depends on lumens, screen gain, room ambient light, and screen size. A 5,000-lumen projector on a 150-inch screen in a bright room may look dim.
"Short-throw projectors are always better for classrooms."	Discuss shadow interference (teacher walking in front), image geometry distortion, and higher cost per lumen. They solve some problems while creating others.
"HDMI is HDMI — all cables carry the same quality signal."	Introduce HDMI versions (1.4, 2.0, 2.1) and bandwidth implications for high-resolution/high-frame-rate content. Explain that passive HDMI cables have length limitations.

3.7 Assessment Alignment Table

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q1-Q2	Resolution and pixel density	Domain 3	2.5	Remember/Understand
Q3	Display technology comparison	Domain 3	2.5	Analyze
Q4	Throw ratio calculation	Domain 3	2.5	Apply
Q5	Luminance and ambient light	Domain 3	2.5	Analyze
Q6-Q7	Viewing distance formula	Domain 3	2.5	Apply

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q8	Color space and calibration	Domain 3	2.5	Understand
Q9	Camera sensor and frame rate	Domain 3	2.5	Understand
Q10	Sight line analysis principles	Domain 3, 8	2.5	Evaluate

3.8 Accommodation Notes

Students who are blind or have low vision will need tactile or verbal descriptions of visual demonstrations. Prepare audio descriptions for all video and image content used in instruction. For throw ratio calculations, provide large-print formula cards. Students with dyscalculia should be provided a step-by-step worked example sheet and allowed extended time on calculation exercises. Color perception demonstrations should note when color-blindness affects the perception of an example — choose demonstration images with sufficient luminance contrast as a secondary cue.

3.9 Chapter Lab Rubric — Lab 3: Projector Throw and Luminance Testing

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All 3 throw calculations correct; formula applied with correct units	2 of 3 calculations correct; minor unit errors	1 of 3 correct; formula misapplied in some cases	Calculations largely incorrect; formula not understood
Procedure	All scenarios attempted; measurement verification performed	All scenarios attempted; verification skipped	2 of 3 scenarios completed	Fewer than 2 scenarios attempted
Documentation	Calculations shown step-by-step;	Calculations shown; brief contextual	Answers given without work shown	Answers only; no work or context

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	results interpreted in context of room design	note		
Teamwork	Work clearly divided; evidence of peer checking and discussion	Generally collaborative; one member slightly dominant	Limited collaboration; parallel rather than cooperative work	No evidence of collaboration

Chapter 4: Signal Flow

4.1 Instructor Overview

Signal flow is the connective tissue of AV systems — understanding it is prerequisite to troubleshooting any system failure. This chapter demands a high level of active learning because the concept does not fully form until students physically trace a signal through actual hardware. Begin with the abstraction (the block diagram) before moving to physical connectors. Students consistently underestimate the diversity of connector types in the AV industry; the connector identification activity is non-negotiable. The cable termination activity requires fine motor manipulation and patience — allow extra time and ensure proper supervision. Key teaching points: the difference between analog and digital signal domains and where conversions occur; the role of signal distribution amplifiers and switchers; and the concept of impedance matching. Troubleshooting methodology (eliminate → isolate → replace) should be introduced here even though Chapter 10 covers it in depth — it is most motivating when students first encounter a signal chain failure they need to solve.

4.2 Lesson Plans

Session A — Signal Domains and Connector Identification

Time	Activity	Details
0-5 min	Warm-Up	Hold up an RJ-45 connector and an XLR connector. Ask: "What is each for? What would happen if you used the wrong one?"
5-25 min	Direct Instruction	Analog vs. digital signal domains. Block diagram reading conventions. Signal sources, processing, distribution, and display. Common connector types: HDMI, DisplayPort, USB-C, SDI, VGA, DVI, XLR, TRS, TS, RJ-45, BNC. Cable vs. connector distinction.
25-40 min	Active Learning	Connector identification card sort. Students match 20 physical connector samples to name, signal type, and typical use. Self-check with answer key.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Draw a simple block diagram of a laptop connected to a projector with audio to a speaker. Label every signal type and connector."

Session B — Distribution, Processing, and Troubleshooting

Introduction

Time	Activity	Details
0-5 min	Warm-Up	Present a scenario: "The projector shows a 'No Signal' message. What are the first three things you check?" Collect answers before instruction.
5-25 min	Direct Instruction	Signal distribution

Time	Activity	Details
		amplifiers (DA), matrix switchers, scalers/converters. HDCP and content protection implications. Impedance matching basics. Analog-to-digital conversion. Introduction to troubleshooting methodology: Eliminate → Isolate → Replace.
25–40 min	Active Learning	Lab 4A: Signal Flow Diagram. Students draw a complete signal flow diagram for the classroom AV system, labeling all signals, connectors, and processing devices.
40–50 min	Debrief / Exit Ticket	Exit ticket: "What is HDCP, and why can it cause problems in educational AV systems? What is one design decision that prevents HDCP-related failures?"

4.3 Discussion Questions

21. A classroom AV system works perfectly with one laptop but shows no signal with another. Using the eliminate-isolate-replace methodology, describe your troubleshooting process step by step.
22. HDCP (High-bandwidth Digital Content Protection) is designed to prevent unauthorized copying of digital content. What are the unintended consequences of HDCP in educational settings, and how should AV designers address them?
23. Why are signal flow diagrams valuable documents even after a system is installed? Who should have access to them, and in what format?
24. Compare the trade-offs between analog and digital signal distribution over long cable runs in a large campus environment. When might one be preferable to the other?

25. The transition from SDI to AV-over-IP for video distribution is accelerating.

What signal flow knowledge remains relevant across both technologies, and what changes fundamentally?

4.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Provide pre-drawn block diagram templates for the lab. Allow students to label a completed diagram before requiring them to draw from scratch. Color-code signal types (red = audio, blue = video, green = data).
Experienced teachers	Ask them to diagram the signal flow of their current classroom system. Many will discover they have never thought about it and will find surprising complexity — or gaps.
AV specialists	Introduce AV-over-IP signal flow concepts (NDI, Dante, AES67) as extension material. Ask them to explain to the class how SDI differs from HDMI in broadcast vs. AV contexts.

4.5 Lab Facilitation Notes

Setup: Assemble a connector sample kit with at least 20 labeled specimens (label the back only — front is for identification). Prepare a signal flow diagram answer template for the classroom system. If cable termination is planned, set up termination stations with RJ-45 crimp tools, keystone jacks, and patch cable materials; verify all crimping tools are in working order before class.

Safety Note: Cable termination involves sharp tools. Brief students on proper tool handling. Ensure anyone with fine motor limitations has an alternative (pre-terminated cable inspection rather than termination).

4.6 Common Student Misconceptions

Misconception	Correction Strategy
"Digital signals don't degrade over distance."	Explain that digital signals do degrade — they are simply more tolerant of degradation until a cliff-edge failure occurs. Introduce the concept of bit error rate and the "digital cliff" vs. analog "snow."
"Any HDMI cable will carry any HDMI signal."	Revisit HDMI versions and bandwidth. Demonstrate or describe what happens when a 4K/60Hz signal is sent over an HDMI 1.4 cable. Discuss premium-certified HDMI for longer runs.
"A matrix switcher and a distribution amplifier are the same thing."	Use a diagram: a DA sends one input to multiple outputs; a matrix switcher can route any input to any output. They solve different distribution problems.
"HDCP problems are the user's fault for using bad cables."	Explain that HDCP handshake failures can occur even with perfect cables when non-HDCP-compliant devices are in the chain. Design, not user behavior, is the correct response.

4.7 Assessment Alignment Table

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q1-Q2	Analog vs. digital signal domains	Domain 4	2.5	Understand
Q3	Connector identification	Domain 4	2.5	Remember/ Apply
Q4	Block diagram reading	Domain 4	2.5	Analyze
Q5	Distribution amplifier function	Domain 4	2.5	Understand
Q6	Matrix switcher function	Domain 4	2.5	Understand
Q7	HDCP and content protection	Domain 4	2.3	Analyze

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q8	Impedance matching basics	Domain 4	2.5	Understand
Q9	Troubleshooting methodology introduction	Domain 4, 10	2.7	Apply
Q10	Signal flow diagram production	Domain 4	2.5	Create

4.8 Accommodation Notes

The cable termination component of Lab 4 requires fine motor dexterity. Offer an alternative for students with motor limitations: inspection and documentation of pre-terminated cables, or use of a cable tester to verify a pre-made cable.

Connector identification card sorts should be available in large-print format.

Diagrams produced in the lab may be completed using digital drawing tools (Lucidchart, draw.io) as an alternative to hand-drawing for students with motor limitations.

4.9 Chapter Lab Rubric — Lab 4: Signal Flow Diagram and Cable Identification

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All connectors correctly identified; signal flow diagram has no errors	≥80% connector accuracy; 1-2 diagram errors	60-79% accuracy; several diagram errors	Below 60% accuracy; diagram fundamentally incorrect
Procedure	All 20 connectors identified; complete diagram produced	All attempted; minor omissions	15-19 connectors; incomplete diagram	Fewer than 15 connectors; no complete diagram
Documentation	Diagram neatly labeled	Diagram labeled;	Partial labeling;	Diagram unlabeled or

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	with signal types, connector names, and direction arrows	direction arrows missing or inconsistent	signal types missing	unreadable
Teamwork	Clear evidence of collaborative discussion on ambiguous connectors	Mostly collaborative	Limited collaboration	No collaboration evident

Chapter 5: Networking for AV

5.1 Instructor Overview

Networking is the chapter that most challenges AV-background instructors and most excites IT-background students. The goal is not to make AV professionals into network engineers — it is to develop enough literacy to collaborate effectively with IT departments and to understand how network decisions affect AV performance. Key teaching points: the OSI model as a framework for understanding where AV-over-IP problems occur; IP addressing and subnetting at a conceptual level (not deep subnetting math); VLANs and QoS as tools for protecting AV traffic; and the specific protocols used in AV-over-IP (Dante, NDI, AVB/TSN, SDVoE). Emphasize the organizational dimension: in educational institutions, AV and IT departments historically operate in silos, and this creates costly integration failures. Teaching students to speak the language of both worlds is a genuine professional differentiator. The firewall and security segment should include a discussion of the increasing vulnerability of network-connected AV systems to cyberattack — this is not theoretical in educational settings.

5.2 Lesson Plans

Session A — IP Fundamentals for AV Professionals

Time	Activity	Details
0–5 min	Warm-Up	Ask: "Has your school's AV system ever been impacted by an IT policy change — software update, firewall rule, VLAN change? What happened?"
5–25 min	Direct Instruction	OSI model overview (focus on layers 1, 2, 3, and 7 for AV). IP addressing: IPv4, subnet masks, DHCP vs. static assignment. MAC addresses. Switches, routers, and access points in AV context. VLANs and network segmentation.
25–40 min	Active Learning	IP address allocation worksheet. Given a school network scenario, students assign static IPs to AV devices, identify subnet conflicts, and propose a VLAN structure. Work in pairs.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Why would an AV engineer request a dedicated VLAN for AV traffic? What problem does it solve?"

Session B — AV-over-IP Protocols and Network Security

Time	Activity	Details
0–5 min	Warm-Up	Show a topology diagram of a modern AV-over-IP system. Ask students to count the number of network-connected

Time	Activity	Details
		devices. Discuss the attack surface this represents.
5-25 min	Direct Instruction	AV-over-IP protocols: Dante (audio), NDI (video), AES67, AVB/TSN, SDVoE. Quality of Service (QoS) configuration for AV. Network security basics: default credential management, firmware updates, network segmentation. BYOD considerations.
25-40 min	Active Learning	Lab 5: Network Map Activity. Students draw a physical and logical network map for the classroom AV system, identifying all network-connected devices, their IP addresses, and their VLAN membership.
40-50 min	Debrief / Exit Ticket	Exit ticket: "List three security vulnerabilities common to network-connected AV systems in schools and one mitigation for each."

5.3 Discussion Questions

26. An IT department insists that all AV devices must use DHCP addressing. An AV integrator insists that AV devices require static IPs. Who is right, and how would you facilitate a productive conversation between these two perspectives?
27. A school discovers that its networked AV system has been using the manufacturer's default username and password since installation three years ago. What risks does this represent, and what remediation steps should be taken in what order?

28. Dante and AVB/TSN both route professional audio over IP networks but have different architectural approaches. In what scenarios would each be preferred in an educational campus environment?
29. BYOD (Bring Your Own Device) policies create network complexity for AV systems. How does a well-designed network architecture protect AV performance while still supporting BYOD?
30. As AV systems become more network-integrated, the boundary between AV support and IT support becomes less clear. How should institutions organize staffing and responsibility to prevent support gaps?

5.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Provide an analogy: IP address = mailing address; MAC address = person's name; router = post office. Use a simple home network as the starting reference before scaling to campus environments.
Experienced teachers	Focus on the organizational dynamics: the AV-IT collaboration problem is fundamentally a communication and governance challenge. Connect to change management concepts they may know from school leadership training.
AV specialists	Assign reading on AVB/TSN (IEEE 802.1) and compare to Dante's proprietary approach. Ask them to present the trade-offs to the class with a recommendation for a hypothetical K-12 campus deployment.

5.5 Lab Facilitation Notes

Setup: Obtain a current network map of the classroom AV system from IT or facilities. If this is unavailable, create a simplified representative diagram. Ensure students have access to the command-line ipconfig (Windows) or ifconfig/ip

(Mac/Linux) tools. A Wi-Fi analyzer app (Wireshark or NetSpot) can supplement the lab if IT approves its use on the network.

Run: Guide students to discover device IP addresses through the control system interface or by logging into network switches (if IT permits). Emphasize that documentation — not configuration — is the lab goal. Students are observers, not network administrators.

5.6 Common Student Misconceptions

Misconception	Correction Strategy
"Wi-Fi is always good enough for professional AV."	Demonstrate latency variation and packet loss under load. Show how 802.11 contention-based access can cause audio dropouts that wired Ethernet would not. Wi-Fi is suitable for control; high-bandwidth uncompressed AV-over-IP generally requires wired connections.
"Network-connected AV devices are secure by default."	Research shows a high percentage of network-connected AV devices in educational institutions use default credentials. Discuss the Shodan search engine and what it reveals about exposed devices.
"NDI and Dante are competing protocols for the same purpose."	Clarify: Dante is primarily a professional audio transport protocol; NDI is primarily a video transport protocol. They are complementary, not competing.
"Adding AV devices to the main school network is the simplest solution."	Explain how AV multicast traffic, particularly from AV-over-IP systems, can saturate a flat network and degrade all network services. VLAN segmentation is a best practice, not optional complexity.

5.7 Assessment Alignment Table

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q1-Q2	OSI model	Domain 5	2.5	Remember/

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
	layers			Understand
Q3	IP addressing basics	Domain 5	2.5	Apply
Q4	DHCP vs. static IP	Domain 5	2.5	Analyze
Q5	VLANs and network segmentation	Domain 5	2.5	Understand
Q6	QoS for AV traffic	Domain 5	2.5	Understand
Q7	Dante protocol	Domain 5	2.5	Understand
Q8	NDI protocol	Domain 5	2.5	Understand
Q9	Network security for AV systems	Domain 5	2.3	Evaluate
Q10	AV-IT collaboration challenges	Domain 5	2.4	Analyze

5.8 Accommodation Notes

Network map diagrams may be completed using accessible digital tools (draw.io, Microsoft Visio) for students who have difficulty with hand-drawn technical diagrams. Ensure all network configuration interfaces used in the lab have keyboard-navigable alternatives to mouse-only UIs. Command-line exercises should include on-screen text magnification for students with low vision. Provide printed reference cards for all CLI commands used.

5.9 Chapter Lab Rubric — Lab 5: Network Map and IP Configuration

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All IP assignments correct; VLAN structure	Minor IP or VLAN errors; overall logic sound	Multiple errors; VLAN concept partially	Significant errors; IP addressing not understood

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	logically sound; no subnet conflicts		understood	
Procedure	All devices identified and mapped; both physical and logical views produced	All devices mapped; one view missing or incomplete	Partial device list; one view only	Incomplete map; missing significant devices
Documentation	Map fully labeled with IPs, VLANs, and device types; legend provided	Most labels present; legend missing	Partial labeling; hard to interpret without context	Unlabeled or uninterpretable
Teamwork	Clear role division; evidence of cross-checking between partners	Mostly collaborative	Limited collaboration	No evidence of joint work

Chapter 6: Control Systems

6.1 Instructor Overview

Control systems represent the user-facing layer of AV infrastructure and the point where technology and pedagogy intersect most directly — a poorly designed control interface can make an excellent AV system effectively unusable by a typical classroom teacher. This chapter should ground students in both the technical architecture of control systems (processors, touch panels, control languages) and the human factors principles that determine whether a system gets used or abandoned. Key teaching points: the difference between centralized and distributed control architectures; the role of the control processor; the logic of macro programming (trigger → condition → action); and the critical importance of user experience design. The lab on macro programming is often the most engaging

activity of the entire course — students feel genuine agency when they successfully automate a sequence of AV actions. Note that with the midterm review in the same week, Session B time must be managed carefully. Do not sacrifice the macro programming activity for review time — use the pre-class period or restructure the review session instead.

6.2 Lesson Plans

Session A — Control System Architecture and Logic

Time	Activity	Details
0-5 min	Warm-Up	Show two touch panel UIs: one cluttered with 30 buttons, one clean with 6. Ask: "Which would a teacher prefer to use in the middle of a lesson? Why?"
5-25 min	Direct Instruction	Control system components: processor, control interfaces (touch panel, button panel, app), feedback devices. Communication protocols: RS-232, RS-485, IP, relay, IR. Centralized vs. distributed architectures. Logic: triggers, conditions, actions. AV system presets and macros.
25-40 min	Active Learning	Logic flow design exercise. Students write a macro flowchart (not code) for "Start Class" and "End Class" system presets for a hypothetical classroom. Share and critique in pairs.
40-50 min	Debrief / Exit Ticket	Exit ticket: "A professor reports that when they press 'Start Class,' the projector turns on but no audio plays. Using your macro flowchart, what are

Time	Activity	Details
		two possible causes?"

Session B — Control Programming Lab + Midterm Review

Time	Activity	Details
0-5 min	Warm-Up	Brief knowledge check: students self-rate confidence (1-5) on each of the six Chapters 1-5 topic areas. Used to target midterm review.
5-20 min	Direct Instruction + Lab	Lab 6: Control macro programming exercise using software simulator or physical panel. Students program a "Present" macro (display on, volume set, lights dim) and test it.
20-35 min	Active Learning	Midterm Review Game (Jeopardy-style or quiz format covering Chapters 1-5). Address the lowest-confidence areas from the warm-up survey.
35-50 min	Debrief / Exit Ticket	Students submit one question they still have about Chapters 1-6 that they want answered before the midterm. Instructor compiles and addresses top questions at start of Week 8.

6.3 Discussion Questions

31. A school district has standardized on a single control system platform for all buildings. A new building requires an AV system that the chosen platform does not support well. How should the district approach this conflict between standardization and technical suitability?

32. User interface design for AV control panels is rarely taught in AV training programs. What principles from UX design apply directly to touch panel design, and what AV-specific considerations are unique to the field?
33. A control system processor is often described as the "brain" of an AV system. What are the risks of centralizing all system functions in a single processor, and how do distributed architectures mitigate these risks?
34. Many teachers report avoiding the installed AV control panel because "it's too complicated" and using their laptop directly instead. What does this tell you about the design process that produced the system, and what would you do differently?
35. Cloud-connected control systems offer remote management and monitoring capabilities but introduce network dependency and potential security vulnerabilities. How should educational institutions weigh these trade-offs?

6.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Focus on the user experience design element of control systems — this is familiar pedagogical territory (designing for the learner). Use flowcharts rather than code for the macro activity.
Experienced teachers	Ask them to evaluate their school's existing control UI against a 5-point usability rubric (clarity, discoverability, feedback, error recovery, efficiency). This produces a real deliverable and validates their perspective.
AV specialists	Allow them to use a real control programming environment (Crestron Simpl, Extron Global Scriptor, or Q-SYS Designer) rather than a simplified simulator. Encourage them to mentor beginners on flowchart logic.

6.5 Lab Facilitation Notes

Setup: If using a software simulator, install and test it on all lab computers at least one week before the session. Common simulators: Q-SYS Designer (free), Crestron Toolbox (free for programming view), Extron Global Configurator (free for logic view). Alternatively, a tablet-based control app (Control4, Roomie Remote) can demonstrate control concepts without programming. Prepare the "Present" and "End Class" macro specifications as a printed handout.

Run: Walk through the macro logic together before students program independently. Emphasize that troubleshooting a non-functioning macro follows the same eliminate-isolate approach as hardware troubleshooting.

Debrief: Have one student demonstrate their macro to the class. Ask: "What would make this UI better for a first-year teacher with no AV training?"

6.6 Common Student Misconceptions

Misconception	Correction Strategy
"Control systems require coding expertise to program."	Distinguish between graphical/flow-based programming environments (Q-SYS, Extron GCP) and text-based programming (Crestron SIMPL+). Many professional control systems are configured using drag-and-drop logic, not traditional coding.
"A more complex control system is always more capable."	Emphasize KISS (Keep It Simple, Stupid) as a professional design principle. Over-engineered systems fail more often and are harder to maintain. Complexity should be justified by a specific user need.
"RS-232 is obsolete and no longer used."	Show current product sheets demonstrating RS-232 ports on professional AV equipment. Explain that many professional AV devices still use serial control because it is deterministic and reliable in ways that network-based control is not.
"The control touchpanel IS the control system."	Clarify: the touch panel is an input device; the control processor is the brain. A control system can function without a touch panel (using sensors, schedulers, or other triggers).

6.7 Assessment Alignment Table

Textbook Review Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's Level
Q1-Q2	Control system components	Domain 6	2.5	Remember/ Understand
Q3	Control communication protocols	Domain 6	2.5	Understand
Q4	Macro logic: trigger-condition-action	Domain 6	2.5	Apply
Q5	Centralized vs. distributed control	Domain 6	2.5	Analyze
Q6	Touch panel UX principles	Domain 6	2.5, 2.6	Evaluate
Q7	Feedback devices in control systems	Domain 6	2.5	Understand
Q8	Cloud control and remote management	Domain 6	2.5, 2.3	Analyze
Q9	Control system troubleshooting	Domain 6, 10	2.7	Apply
Q10	System presets and standardization	Domain 6	2.5	Evaluate

6.8 Accommodation Notes

Touch panel interfaces used in the lab must be keyboard-navigable or have a software equivalent accessible to students who cannot use touch input. Provide printed flowchart templates as an alternative to digital diagramming tools for the

macro logic exercise. Ensure the control simulator software meets WCAG 2.1 AA standards or provide an alternative text-based macro description format. For students with cognitive load challenges, provide the "Present" macro as a partially completed flowchart to complete rather than creating from scratch.

6.9 Chapter Lab Rubric — Lab 6: Control Panel Macro Programming

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	Macro executes correctly; all devices respond in correct sequence; no logic errors	Macro mostly functional; minor sequencing issue	Macro partially functional; logic errors present	Macro non-functional; logic fundamentally flawed
Procedure	Both "Present" and "End Class" macros completed and tested	Both attempted; one untested	One macro completed	No functional macro produced
Documentation	Flowchart complete, legible, and matches programmed logic exactly	Flowchart present with minor discrepancies	Flowchart incomplete or significantly different from program	No flowchart produced
Teamwork	Both partners actively debugged; evidence of shared problem-solving	Both engaged; one partner more active in debugging	One partner primarily responsible	No collaborative work evident

① Note to Instructor — Chapters 7–12

The following chapter notes follow the same structure as Chapters 1–6 (Overview, Lesson Plans, Discussion Questions, Differentiation, Lab Notes, Misconceptions,

Assessment Alignment, Accommodations, Rubric). Due to the guide's length, each is presented in its full essential form.

Chapter 7: Collaboration Technology

7.1 Instructor Overview

Chapter 7 arrives in the second half of the course and directly addresses the most visible AV challenge in post-pandemic education: hybrid learning environments. Students come to this chapter with intense personal experience as both frustrated end users and struggling facilitators of hybrid meetings. Leverage that experience. The key distinction to establish early is between synchronous video conferencing (Zoom, Teams, Meet — software-based UC&C) and room-based collaboration systems (dedicated hardware endpoints, codec-based systems). Students frequently assume that Zoom on a laptop is equivalent to a properly designed room system — and it is not. Spend meaningful time on the equity dimension of hybrid: remote participants in a poorly designed hybrid room receive a fundamentally unequal educational experience. Camera placement, microphone coverage, and acoustic design all affect whose voice and perspective is captured and transmitted. The BYOD section should include a practical wireless presentation system comparison (Mersive Solstice, Barco ClickShare, AirPlay, Miracast) with clear evaluation criteria.

7.2 Lesson Plans

Session A — UC&C Platforms and Room System Architecture

Time	Activity	Details
0-5 min	Warm-Up	"Raise your hand if you've been a remote participant in a meeting where you felt like a second-class attendee. What made it feel that way?"

Time	Activity	Details
5-25 min	Direct Instruction	UC&C ecosystem: Zoom, Teams, Google Meet, Webex. Hardware room systems vs. software-only solutions. Camera types: PTZ, fixed-angle, intelligent tracking, 360-degree. Microphone coverage for hybrid rooms. Codec architecture.
25-40 min	Active Learning	Hybrid room evaluation: students use a rubric to assess the classroom's existing collaboration setup from the perspective of a remote participant. Score and compare.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Name the two most significant improvements you would make to this classroom's hybrid capability for under \$5,000."

Session B — BYOD, Wireless Presentation, and Hybrid Equity

Time	Activity	Details
0-5 min	Warm-Up	Quick demo: connect a phone via Miracast, AirPlay, and a wired HDMI dongle. Note which works first and why.
5-25 min	Direct Instruction	BYOD wireless presentation: Miracast, AirPlay, Chromecast, Mersive Solstice, Barco ClickShare. Trade-offs in reliability, security, and network load. Hybrid equity framework: camera placement, microphone array coverage, acoustic

Time	Activity	Details
		treatment. AI-enhanced camera tracking.
25–40 min	Active Learning	Lab 7: Hybrid Room Setup and Test. Students configure a video conferencing session in the room, evaluate remote participant experience from a second device, and document findings.
40–50 min	Debrief / Exit Ticket	Exit ticket: "What is the single most important investment a school could make to improve hybrid equity for remote learners, and how would you justify that cost to an administrator?"

7.3 Discussion Questions

36. How does the camera placement in a hybrid classroom affect whose perspective dominates the remote experience — the teacher's or the students'? What are the pedagogical implications?
37. A school district cannot afford dedicated room systems and relies on teacher-owned laptops for all video conferencing. What low-cost interventions can meaningfully improve the hybrid experience?
38. AI-powered camera tracking has become a common feature in collaboration systems. What are the benefits and the risks — including privacy, bias in object detection, and instructor autonomy?
39. In a hybrid class, a student's home environment (noisy, poorly lit, limited bandwidth) affects their participation quality. How is this an equity issue, and what responsibilities does the institution have?
40. Compare synchronous video conferencing to asynchronous video communication (recorded lectures, video discussion boards) as collaboration

modalities. When is each more appropriate, and how does AV design differ for each?

7.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Start with the student's own Zoom/Teams experience and build up. Focus on the 3 most impactful improvements: camera angle, microphone, and lighting.
Experienced teachers	Ask them to redesign their own class's hybrid setup based on what they learn in this chapter. Connect to their existing pedagogy — how does hybrid change facilitation decisions?
AV specialists	Assign advanced reading on intelligent tracking camera algorithms and their limitations. Challenge them to design a hybrid system specification for a 50-person lecture hall.

7.5 Lab Facilitation Notes

Ensure a second device (tablet or laptop) is available for students to evaluate the remote participant experience. Run the VC session at full quality — do not use a low-bandwidth test. Document remote participant experience across: video quality, audio clarity, ability to see whiteboard, ability to see other participants' faces, ability to ask questions. Compare experiences across groups.

7.6 Common Student Misconceptions

Misconception	Correction Strategy
"A good internet connection is all you need for good hybrid teaching."	Demonstrate the same connection with poor camera angle, reverberant audio, and backlighting to show that bandwidth is only one variable.
"Wireless presentation is always less reliable than wired."	Compare enterprise wireless presentation systems to consumer-grade solutions. Modern enterprise systems (Solstice, ClickShare) offer

Misconception	Correction Strategy
	enterprise-grade reliability — the problem is mismatched tools, not the technology category.
"AI camera tracking eliminates the need for thoughtful camera placement."	Demonstrate AI tracking failure cases: large groups, rapid room movement, instructor standing near students. Physical placement remains the primary design decision; AI enhances but does not replace it.

7.7 Assessment Alignment Table

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
Q1-Q2	UC&C platforms and hardware endpoints	Domain 7	2.4, 2.6	Understand
Q3-Q4	Camera types and placement	Domain 7	2.5	Apply/Analyze
Q5	Microphone coverage in hybrid rooms	Domain 7	2.5	Analyze
Q6	BYOD wireless presentation	Domain 7	2.5	Analyze
Q7	Hybrid equity principles	Domain 7	2.3, 2.6	Evaluate
Q8	AI camera tracking trade-offs	Domain 7	2.7	Evaluate
Q9	Bandwidth requirements	Domain 5, 7	2.5	Apply
Q10	Remote participant experience design	Domain 7	2.5, 2.3	Create

7.8 Accommodation Notes

Captioning must be enabled on all video conferencing demonstrations. If using a platform without automatic captions, arrange a CART provider. Students who are Deaf or hard of hearing should evaluate the captioning quality as part of the lab assessment — this is a valid professional competency. Ensure remote participant evaluation can be performed by students using screen readers (confirm compatibility with the VC platform's accessibility features).

7.9 Chapter Lab Rubric — Lab 7: Hybrid Meeting Room Setup and Test

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	System configured correctly; all components functional; remote view excellent	Mostly correct; one component issue	Partial configuration; significant issues	System non-functional or unconfigured
Procedure	Full setup-test-document cycle completed	Setup and test complete; documentation brief	Partial setup only	No functional setup
Documentation	Remote participant evaluation report complete with specific metrics and recommendations	Report present; recommendations general	Report sparse; no recommendations	No report produced
Teamwork	Both in-room and remote roles played by all group members	Roles shared; one perspective underrepresented	One member dominated setup	No equitable role distribution

Chapter 8: Classroom AV Design

8.1 Instructor Overview

Chapter 8 is the synthesis chapter of the first half of the course — it brings audio (Ch. 2), video (Ch. 3), signal flow (Ch. 4), networking (Ch. 5), and control (Ch. 6) together into the design of a complete room. This is where abstract principles become concrete professional practice. The key reference standard is ANSI/INFOCOMM 1M:2009 (Audiovisual Systems Performance Verification) and the ANSI/AVIXA 2M:2010 (Display Image Size for 2-D Content in Audiovisual Systems). Instructors should familiarize themselves with these documents before teaching this chapter. The design exercise (Lab 8) is the most complex lab in the course — allow students to struggle productively with the trade-offs rather than providing immediate answers. A classroom with good audio coverage may require speaker placement that conflicts with the structural realities of the room; this is realistic professional tension. Connect explicitly to the capstone project: the AV Upgrade Plan students will develop for the final is essentially an application of what they learn in this chapter.

8.2 Lesson Plans

Session A — Room Geometry, Sight Lines, and Audio Coverage

Time	Activity	Details
0–5 min	Warm-Up	Provide a blank room floor plan. Ask students to mark where they would sit and where they would place: a display, a speaker, and a camera. Compare results.
5–25 min	Direct Instruction	Sight line analysis: ANSI/AVIXA 2M viewing distance criteria (1.5H–6H rule). Display placement: tilt, height, horizontal offset. Speaker coverage: room mode, delay,

Time	Activity	Details
		distributed vs. point-source systems. Audience area determination.
25-40 min	Active Learning	Sight line calculation exercise. Given a room plan with dimensions, students calculate maximum viewing distance, minimum display height, and identify seating zones with acceptable sightlines.
40-50 min	Debrief / Exit Ticket	Exit ticket: "A room is 60 feet deep and the screen height is 8 feet. What is the maximum viewing distance from the screen recommended by ANSI/AVIXA 2M?"

Session B — System Specification and the Design Process

Time	Activity	Details
0-5 min	Warm-Up	Present a one-page "Design Brief" for a hypothetical classroom. Ask: "What information is missing that you would need before selecting any equipment?"
5-25 min	Direct Instruction	AV design process: programming (needs assessment), schematic design, design development, construction documents, commissioning. Equipment selection criteria: performance, cost, maintainability, lead time. Rack design basics. Cable management principles.
25-40 min	Active Learning	Lab 8: Classroom AV

Time	Activity	Details
		Design Exercise. Students produce a one-page AV design for a provided room plan including: display specification, speaker layout, signal flow diagram, and basic equipment list.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Your design has been rejected by the building's architect because your speaker placement requires cutting into a structural beam. What are two alternative solutions?"

8.3 Discussion Questions

41. A high school is renovating 20 classrooms and asks you to recommend a standard AV package for all rooms. What are the benefits and risks of classroom standardization from a design, maintenance, and pedagogical flexibility perspective?
42. The ANSI/AVIXA sight-line standards were developed primarily for traditional front-of-room lecture formats. How do they need to be adapted for collaborative, flexible, or active-learning classroom configurations?
43. A community college classroom built in 1975 has structural and acoustic limitations that make meeting current AV design standards impossible without significant renovation. How do you design the best possible system within real-world constraints?
44. Who should be involved in the AV programming phase of a classroom design project — and who is typically left out? What happens when end users (teachers, students with disabilities) are not consulted?
45. Sustainable design is increasingly prioritized in educational construction. How can AV system design contribute to (or detract from) a building's energy efficiency and environmental goals?

8.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Provide a partially completed design template. Focus on the sight-line calculation as the entry point — it is arithmetic-based and builds confidence before the more complex audio coverage element.
Experienced teachers	Use their actual classroom as the design site. They have expert knowledge of how the room is used — this is a genuine data asset in the programming phase. Connect explicitly to the capstone project.
AV specialists	Require them to produce a complete signal flow diagram, equipment list with part numbers (vendor-agnostic categories), and a basic rack layout as an extension to the standard lab deliverable.

8.5 Lab Facilitation Notes

Provide three room plan options of varying difficulty (simple rectangular classroom, L-shaped open learning space, tiered lecture hall). Allow students to choose their challenge level. Have ANSI/AVIXA 2M tables available for reference. A flexible drawing tool (graph paper, or draw.io) works well for the design exercise. Circulate and ask questions rather than giving answers: "What criterion are you using to make that speaker placement decision?"

8.6 Common Student Misconceptions

Misconception	Correction Strategy
"AV design begins with selecting equipment."	Emphasize the programming phase: needs assessment, room survey, and user consultation must precede any equipment selection. Selecting equipment first is the most common cause of costly redesigns.
"Standards tell you exactly what to	Standards set minimum performance

Misconception	Correction Strategy
install."	thresholds — they tell you what the system must achieve, not how to achieve it. Design judgment fills the gap between the standard and the specific solution.
"A distributed speaker system always sounds better in classrooms."	Distributed systems provide more even coverage, but a well-tuned point-source system in a well-treated room can achieve excellent intelligibility. Context determines the best solution.

8.7 Assessment Alignment Table

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
Q1-Q2	ANSI/AVIXA 2M viewing distance criteria	Domain 8	2.5	Apply
Q3	Sight line analysis calculation	Domain 8	2.5	Apply
Q4-Q5	Speaker coverage and placement	Domain 8	2.5	Analyze
Q6	AV design process phases	Domain 8	2.5	Understand
Q7	Equipment selection criteria	Domain 8	2.5, 2.7	Evaluate
Q8	Rack design basics	Domain 8	2.5	Understand
Q9	Stakeholder involvement in design	Domain 8	2.4	Evaluate
Q10	Sustainable AV design principles	Domain 8	2.3	Evaluate

8.8 Accommodation Notes

Room design floor plan exercises should be available in digital format for screen-reader use; provide coordinate-based descriptions of room dimensions as an alternative to graphical floor plans. For students with fine motor limitations, digital drawing tools should be offered. All ANSI standard reference tables should be provided in accessible PDF format with proper heading structure and alt text for any included figures.

8.9 Chapter Lab Rubric — Lab 8: Classroom AV Design Exercise

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	Sight lines, coverage, and signal flow all meet ANSI criteria; no errors	Meets criteria with 1–2 minor deviations	Several criteria not met; explanation provided	Criteria largely unmet; no evidence of standard applied
Procedure	Complete design: display, audio, signal flow, equipment list	All components present; one area underdeveloped	2 of 4 components complete	Fewer than 2 components
Documentation	Design drawing clearly annotated; equipment list with specifications	Drawing annotated; list present but incomplete	Drawing present but minimal annotation	No drawing or list
Teamwork	Design decisions clearly collaborative; conflicting trade-offs resolved through discussion	Mostly collaborative; some unilateral decisions	Limited collaboration	No collaborative evidence

Chapter 9: Digital Signage

9.1 Instructor Overview

Digital signage is the AV application most visible to students, faculty, and visitors on campus — and one of the most frequently misused. This chapter should reframe digital signage from "moving wallpaper" to a strategic communication and wayfinding tool with genuine pedagogical and safety applications. Key teaching points: the difference between media player, content management system (CMS), and display as distinct components; content design principles (dwell time, reading distance, contrast); and accessibility requirements for public digital signage (ADA, WCAG 2.1 Level AA). The CMS scheduling lab is accessible to students with varying technical backgrounds and often produces the highest engagement of any Chapter 9 activity. Use real campus signage as case studies — students have strong opinions about what works and what doesn't, and those opinions are anchored in daily experience. Push back on "flashier is better" — the evidence on effective campus signage favors simplicity, high contrast, and appropriate dwell times.

9.2 Lesson Plans

Session A — Digital Signage Architecture and Content Principles

Time	Activity	Details
0-5 min	Warm-Up	Show 3 images of campus digital signs: one effective, one cluttered, one inaccessible (low contrast). Ask students to rank and explain.
5-25 min	Direct Instruction	Digital signage components: media player hardware, CMS, display, network. Content design principles: dwell time (8-10 seconds per zone), reading distance formula, font size for viewing distance, contrast

Time	Activity	Details
		ratios, animation vs. static. CMS architectures: cloud vs. on-premise.
25-40 min	Active Learning	Content audit: students photograph and evaluate 5 campus digital signs against a provided rubric (contrast, dwell, font size, accessibility, relevance). Score and compare.
40-50 min	Debrief / Exit Ticket	Exit ticket: "The viewing distance to a digital sign is 15 feet. Using the standard formula, what is the minimum font size recommended for body text?"

Session B — CMS Scheduling, Accessibility, and Emergency Messaging

Time	Activity	Details
0-5 min	Warm-Up	Ask: "Has your campus ever used digital signage for an emergency notification? How effective was it? What were its limitations?"
5-25 min	Direct Instruction	CMS scheduling: zones, playlists, dayparting, conditional triggers. Digital signage accessibility: WCAG 2.1 for dynamic displays, captions for video content, screen reader inaccessibility of touch kiosks. Emergency alert system (EAS) integration. ADA requirements for interactive signage.
25-40 min	Active Learning	Lab 9: Content creation and scheduling exercise using a CMS trial account.

Time	Activity	Details
		Students create a 3-slide playlist with correct dwell times, evaluate contrast ratios, and schedule for a campus building.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Name two WCAG 2.1 success criteria that apply to campus digital signage and explain how you applied each in today's lab."

9.3 Discussion Questions

46. Digital signage is increasingly used for emergency notifications on campuses. What are the design requirements for an effective mass-notification display system, and how does this differ from everyday informational signage?
47. A campus's digital signage network is managed centrally by marketing, but departments want control over their building displays. How should governance and content permissions be structured to serve both needs?
48. WCAG 2.1 was not designed with dynamic display signage in mind. What are the limitations of applying web accessibility guidelines to digital signage, and what supplemental standards or practices fill the gap?
49. Data-driven digital signage (displaying real-time information like room availability, dining menus, or bus schedules) requires integration with campus data systems. What data governance and privacy questions should be addressed before implementing such systems?
50. Research on campus digital signage effectiveness suggests that many systems are largely ignored by the audiences they are meant to serve. What design and placement decisions most strongly predict whether a sign is noticed and read?

9.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Use familiar consumer tools (Google Slides, Canva) as analogs to professional CMS interfaces before introducing commercial platforms. Focus on content design principles before system architecture.
Experienced teachers	Connect digital signage to classroom display use: the same contrast, font size, and dwell principles that make a campus sign effective apply to projected instructional slides.
AV specialists	Assign advanced topics: CMS API integration with campus data systems, digital signage over AV-over-IP infrastructure, SoC (System-on-Chip) display technology. Ask them to present a CMS comparison to the class.

9.5 Lab Facilitation Notes

Use a free-tier CMS account (Xibo, Screenly, or Yodeck all offer free tiers). Set up accounts before class and test upload, scheduling, and playback functions. Prepare a content brief specifying what each slide should communicate, so students focus on design and accessibility rather than content ideation. Have a color contrast checker tool ready (WebAIM Contrast Checker works well for planning content).

9.6 Common Student Misconceptions

Misconception	Correction Strategy
"Animated content is more engaging and therefore better."	Present research: animation can increase initial notice but decreases readability and increases cognitive load for complex information. Use animation purposefully — for directional cues, not decoration.
"Digital signage doesn't need to meet accessibility standards because you can just read the physical sign instead."	ADA requirements apply to all public-facing information systems; where digital signage is the primary information channel (emergency alerts, directory listings), it must meet accessibility standards. Interactive kiosks face particularly stringent

Misconception	Correction Strategy
	requirements.
"A more expensive media player produces better image quality."	Display quality depends primarily on the display panel and content resolution, not the media player. Player selection is about reliability, CMS compatibility, and connectivity — not raw image processing.

9.7 Assessment Alignment Table

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
Q1-Q2	Digital signage components	Domain 9	2.5	Remember/Understand
Q3	Dwell time and content design	Domain 9	2.5	Apply
Q4	CMS scheduling and zones	Domain 9	2.5	Apply
Q5	Accessibility for digital signage	Domain 9, 11	2.3	Evaluate
Q6	Emergency alert integration	Domain 9	2.5, 2.3	Analyze
Q7	Font size for viewing distance	Domain 9	2.5	Apply
Q8	CMS governance and permissions	Domain 9	2.4	Evaluate
Q9	Interactive kiosk ADA requirements	Domain 9, 11	2.3	Analyze
Q10	Content effectiveness research	Domain 9	2.7	Evaluate

9.8 Accommodation Notes

Color contrast activities should be conducted using both visual evaluation and quantitative tool measurement, as some students cannot reliably judge contrast visually. Ensure CMS lab tools are keyboard-accessible. Discuss the specific accessibility barriers that digital signage poses for users who are Blind (content is entirely visual) and what the obligations are — this is a real professional and ethical issue, not merely a compliance checkbox.

9.9 Chapter Lab Rubric — Lab 9: Digital Signage Content Creation and Scheduling

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All slides meet contrast, dwell, and font-size criteria; scheduling correctly set	Minor deviations from criteria; scheduling functional	Multiple criteria not met; scheduling partially complete	Criteria largely unmet; scheduling not completed
Procedure	3-slide playlist created, uploaded, scheduled, and verified	3 slides created and scheduled; verification step skipped	2 slides created; scheduling incomplete	1 or no slides; not scheduled
Documentation	Accessibility evaluation form complete with specific WCAG criteria cited	Form complete; WCAG citations vague	Form partially complete; no WCAG reference	No documentation
Teamwork	Design decisions shared; both partners contributed to content and evaluation	Mostly shared; one area dominated	Limited collaboration	No collaboration

Chapter 10: Maintenance and Troubleshooting

10.1 Instructor Overview

Chapter 10 is the most practically applicable chapter for K-12 technology coordinators and facilities staff, who are often first responders to AV system failures. The troubleshooting methodology introduced in Chapter 4 is now formalized: Define the problem → Gather information → Identify possible causes → Test hypotheses → Implement solution → Verify → Document. Instructors should emphasize documentation as a professional imperative, not an afterthought — the service record of a system is as valuable as the system itself for long-term management and warranty support. Preventive maintenance is the most neglected area in educational AV: projector filters go uncleaned, display calibration drifts, software updates accumulate, and cable connections corrode. The PM checklist lab should feel like real professional work because it is. Key equity note: schools with dedicated AV/IT staff maintain systems more reliably; under-resourced schools with no dedicated support staff have poorer performing AV on average — and their students bear the educational cost of that disparity.

10.2 Lesson Plans

Session A — Systematic Troubleshooting Methodology

Time	Activity	Details
0-5 min	Warm-Up	Present a fault scenario: "The projector shows a blue screen. No video. Audio is working." Ask for first steps. Collect responses without judgment.
5-25 min	Direct Instruction	Troubleshooting methodology: 7-step process. Common AV fault categories: power, signal, control, display, audio, network. Systematic fault isolation.

Time	Activity	Details
		Test equipment: cable tester, signal generator, multimeter basics for AV. When to escalate vs. attempt repair.
25-40 min	Active Learning	Fault simulation exercise. Instructor pre-introduces a fault into the classroom system (disconnected input, wrong input selected, muted audio, etc.). Students work through the 7-step process to identify and resolve the fault.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Write a service report for the fault you resolved today. Include: problem statement, steps taken, root cause, resolution, and prevention recommendation."

Session B — Preventive Maintenance and Documentation

Time	Activity	Details
0-5 min	Warm-Up	Ask: "When was the last time the AV system in this room had a preventive maintenance inspection?" (Likely unknown.) Discuss what this means for reliability.
5-25 min	Direct Instruction	Preventive maintenance categories: cleaning (projector filters, lens, vents), firmware updates, lamp life management, battery replacement (wireless mics, remotes), cable inspection, display calibration, control system testing. Service documentation: work

Time	Activity	Details
		order systems, as-built drawings, equipment schedules, warranty records.
25-40 min	Active Learning	Lab 10: PM Checklist. Students execute a 20-point PM checklist on the classroom AV system, documenting findings and flagging items requiring escalation.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Based on today's PM inspection, what is the single highest-priority maintenance action for this system, and what is the consequence of deferring it?"

10.3 Discussion Questions

51. Many educational institutions have no formal AV preventive maintenance schedule. What are the systemic reasons for this, and what would a compelling case for a funded PM program include?
52. When an AV system fails during a class or meeting, who should be called and how quickly should they respond? What does an effective AV support response protocol look like at an institution level?
53. Reactive maintenance (fixing things when they break) vs. preventive maintenance (maintaining before failure): compare the total cost of ownership implications for a school district with 50 classrooms.
54. Service documentation (as-built drawings, equipment schedules, warranty records) is often lost when a vendor or integrator disengages. What contractual provisions should educational institutions include in AV installation contracts to protect access to this documentation?

55. How does the troubleshooting methodology in this chapter apply beyond AV — to other technology systems in an educational institution, or to pedagogical problem-solving?

10.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Provide a laminated "First Responder Checklist" (power → input → cable → volume → connection) that covers 80% of common classroom AV failures with no specialized knowledge required.
Experienced teachers	Ask them to write a brief "AV SOS" protocol for their school — who to call, what information to have ready, what they can try themselves. This is directly deployable professional work.
AV specialists	Assign advanced topics: SNMP monitoring for AV devices, control system diagnostics, predictive maintenance using device telemetry. Ask them to design a monitoring dashboard concept for a 20-room campus.

10.5 Lab Facilitation Notes

Prepare the PM checklist in advance for the specific equipment in your classroom. The checklist should include items that are actually actionable (projector filter inspection, cable connection check, firmware version check) and items that require escalation (lamp replacement if beyond rated hours, structural mounts requiring a technician). Introduce at least one real fault for the fault simulation — do not simulate by telling students what is wrong; let them find it. Pre-testing the PM checklist against your actual system takes 30–45 minutes and is essential preparation.

10.6 Common Student Misconceptions

Misconception	Correction Strategy
"If it ain't broke, don't fix it — preventive maintenance is unnecessary overhead."	Present the data: projector lamp failures are the #1 cause of AV system downtime in educational settings, and they are almost entirely preventable with lamp hour tracking and scheduled replacement before failure.
"Troubleshooting is a talent, not a skill — some people are just good at it."	Demonstrate that troubleshooting is a systematic methodology anyone can learn. The 7-step process works for people with no natural intuition — it structures the thinking so intuition is not required.
"Rebooting solves most AV problems."	While rebooting is a valid first step, it is not a solution — it is a restart. If the problem recurs, rebooting has merely deferred diagnosis. Encourage documenting what conditions preceded the failure before rebooting.
"Firmware updates should be applied as soon as they are released."	Explain the value of testing firmware updates in a non-production environment first, especially for control system firmware that could change device behavior. Read release notes before updating.

10.7 Assessment Alignment Table

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
Q1-Q2	Troubleshooting 7-step methodology	Domain 10	2.7	Understand/Apply
Q3	Common AV fault categories	Domain 10	2.7	Analyze
Q4	PM categories and schedule	Domain 10	2.7	Apply
Q5	Lamp life and lamp management	Domain 10	2.7	Apply
Q6	Service documentation	Domain 10	2.7	Understand

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
	n types			
Q7	When to escalate a fault	Domain 10	2.7	Analyze
Q8	Firmware update best practices	Domain 10	2.7	Apply
Q9	Reactive vs. preventive maintenance cost	Domain 10	2.7	Evaluate
Q10	Support response protocol design	Domain 10	2.4	Create

10.8 Accommodation Notes

PM checklist tasks involving physical climbing (checking ceiling-mounted projectors) or fine manipulation (cable connector inspection) should have an alternative documentation task for students with relevant physical limitations. Ensure service report writing can be completed via voice dictation or digital typing as alternatives to handwriting. Fault simulation scenarios should be documented in text so students with processing speed challenges can reference them during the exercise.

10.9 Chapter Lab Rubric — Lab 10: PM Checklist and Fault Simulation

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	Fault correctly identified and resolved; PM items assessed accurately	Fault resolved; 1-2 PM items assessed incorrectly	Fault partially resolved; PM mostly completed	Fault not resolved; PM incomplete
Procedure	7-step methodology followed	Methodology followed; 18-19 PM items	Methodology partially followed; 14-	Methodology not followed; fewer than 14

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	explicitly; all 20 PM items addressed	addressed	17 PM items	PM items
Documentation	Service report complete with root cause, resolution, and prevention recommendation	Report complete; prevention recommendation vague	Report incomplete; root cause not identified	No report or no usable information
Teamwork	Clear role rotation; all members engaged in both fault and PM activities	Mostly shared; one area dominated	One member led; others assisted minimally	No evidence of collaboration

Chapter 11: Accessibility in AV

11.1 Instructor Overview

Chapter 11 is the equity centerpiece of the course and deserves the full week it is allotted. This chapter should not be framed as compliance — it should be framed as design justice. The core argument: AV systems that exclude users with disabilities are not just legally problematic, they are professionally and ethically unacceptable. Key teaching points: ADA Title II requirements for public educational institutions; Section 508 for federally funded institutions; WCAG 2.1 for digital AV interfaces; ANSI/ASA S12.60 for classroom acoustics and hearing; and the specific AV features that address the needs of users who are Deaf/hard of hearing, blind/low vision, and those with motor, cognitive, and speech disabilities. Students who have themselves navigated inaccessible AV systems — either as students or instructors — often provide the most powerful testimony in this chapter. Create space for that experience to be shared. The accessibility audit lab (Lab 11) is one of the most professionally valuable activities in the course because it produces a real deliverable: a documented assessment of an actual campus space.

11.2 Lesson Plans

Session A — Legal Framework and Disability-Specific AV Features

Time	Activity	Details
0–5 min	Warm-Up	Show a brief video clip without captions and with no audio description. Ask: "What was communicated in that clip to someone who is Deaf? To someone who is Blind?" Discuss what was lost.
5–25 min	Direct Instruction	Legal framework: ADA Title II (public entities), Section 508 (federal), AIA/ANSI A117.1, WCAG 2.1. Disability categories and AV implications: Deaf/HoH (captioning, inductive loop/hearing loop, visual notification), Blind/low vision (audio description, accessible interfaces), motor disabilities (remote control accessibility, switch access), cognitive disabilities (simplified UI, consistent design).
25–40 min	Active Learning	AV accessibility mapping exercise. Students map which accessibility features address which disability categories. Then identify which features are present (or absent) in the current classroom.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Name one ADA requirement that applies to classroom AV systems that is frequently unmet in K–12 schools, and explain what remediation would look like."

Session B — Hearing Loops, Audio Description, and the Accessibility Audit

Time	Activity	Details
0-5 min	Warm-Up	Demonstrate an assistive listening device (hearing loop or FM system) if available. If not, show a video demonstrating the difference in audio quality for a hearing aid user with and without a loop system.
5-25 min	Direct Instruction	Induction loop (hearing loop) technology: how it works, IEC 60118-4 standard, ADA requirements for large areas. Audio description standards. Accessible remote control design. Captioning requirements for pre-recorded vs. live content. Universal Design for Learning (UDL) principles as AV design framework.
25-40 min	Active Learning	Lab 11: Accessibility Audit. Students conduct a structured accessibility audit of a campus AV space using the AVIXA/ISTE accessibility audit form (provided). Identify gaps and prioritize recommendations.
40-50 min	Debrief / Exit Ticket	Exit ticket: "Present one finding from your accessibility audit. What is the gap? What standard applies? What is the remediation, and what does it cost to implement?"

11.3 Discussion Questions

56. The social model of disability argues that it is the environment, not the person, that creates disability. How does this model apply to AV system design in educational spaces?
57. Many educational institutions treat accessibility as a retrofit — adding accommodations after a system is designed and installed. What does accessibility-first or universal design look like in AV system procurement and design processes?
58. Automatic speech recognition (ASR) captioning has dramatically improved in accuracy but still performs poorly for accented speech, technical vocabulary, and multiple simultaneous speakers. What are the implications for relying on AI captions as an accessibility solution?
59. Hearing loop systems are required by the ADA for large assembly areas but are rarely installed in classrooms. What are the barriers to broader implementation, and what would a school district need to do to change that?
60. A student who is blind is enrolled in your AV course. How do you adapt the labs — connector identification, projector measurement, network mapping — to be fully inclusive? Which adaptations benefit all students?

11.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Use the "curb cut effect" as an entry point: accessibility features designed for one group frequently benefit everyone (captions help in noisy environments; high contrast helps in bright rooms). This reframes accessibility from burden to asset.
Experienced teachers	Connect to their legal obligations as educators under IDEA, Section 504, and ADA. Ask them to evaluate their own teaching materials and classroom for AV accessibility gaps.
AV specialists	Assign the IEC 60118-4 hearing loop

Learner Profile	Strategy
	performance standard and ask them to evaluate whether the auditorium or main assembly space on campus meets its requirements. This is specialist-level work with immediate institutional value.

11.5 Lab Facilitation Notes

The accessibility audit form should cover the following categories: Hearing access (loop system, captioning, hearing device compatibility), Visual access (display height, viewing distance, font size, audio description), Physical access (reach, clearance, operable controls), and Digital interface access (screen reader compatibility, keyboard navigation). Assign different audit spaces to different groups to produce a broader campus picture. Brief relevant staff (library, facilities) about the student audit visits in advance.

11.6 Common Student Misconceptions

Misconception	Correction Strategy
"Accessibility is for a small percentage of users and not worth the cost."	Present data: approximately 26% of US adults have some form of disability. In educational settings, the percentage of students requiring accessibility accommodations is often higher. Accessibility features also benefit non-disabled users situationally.
"Captioning is only necessary for students who are Deaf."	Research shows captions improve comprehension for non-native language speakers, students with attention differences, and all learners in noisy environments. Caption accuracy and availability is a universal quality issue.
"If we provide an accommodation on request, we are compliant."	ADA Title II generally requires proactive compliance — systems should be accessible by design, not only when an accommodation is requested. Reactive accommodation is often inferior in quality and dignity-respecting.
"Accessibility and good design are in tension."	Present examples of accessible AV systems that have won industry design awards. Introduce the principle that the

Misconception	Correction Strategy
	best designs succeed for the widest range of users — accessible design is good design.

11.7 Assessment Alignment Table

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
Q1-Q2	ADA and Section 508 requirements	Domain 11	2.3	Understand
Q3	Hearing loop technology (IEC 60118-4)	Domain 11	2.3, 2.5	Understand
Q4	Captioning requirements (live vs. pre-recorded)	Domain 11	2.3	Apply
Q5	WCAG 2.1 for AV interfaces	Domain 11	2.3	Apply
Q6	UDL principles in AV design	Domain 11	2.3, 2.5	Analyze
Q7	Audio description standards	Domain 11	2.3	Understand
Q8	Accessibility audit methodology	Domain 11	2.7	Apply
Q9	Social model of disability in design	Domain 11	2.3	Analyze
Q10	Remediation prioritization and cost	Domain 11	2.7	Evaluate

11.8 Accommodation Notes

The audit itself should model the accessibility principles being taught. Ensure all materials, including the audit form, are available in accessible formats before this chapter. If students with disabilities are in the cohort, invite (but do not require)

them to share relevant lived experience. Never position a student with a disability as a "representative" of their disability category or as a teaching resource without their explicit, ongoing consent.

11.9 Chapter Lab Rubric — Lab 11: Accessibility Audit of a Campus AV Space

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All gaps correctly identified; applicable standards cited accurately for each gap	Most gaps identified; 1-2 standard citations incorrect	Some gaps identified; standards cited vaguely	Few gaps identified; no standards cited
Procedure	All four audit categories fully evaluated; scoring applied consistently	All categories evaluated; scoring minor inconsistencies	3 of 4 categories evaluated	Fewer than 3 categories; no scoring
Documentation	Written report with prioritized recommendations, cost estimates, and standard citations	Report present; prioritization without cost estimates	Findings listed without prioritization or recommendations	No usable written report
Teamwork	All group members audited different categories; collaborative report writing evident	Most categories shared; one member dominant in writing	Limited collaboration; one member did most work	No collaboration

Chapter 12: Emerging Technologies

12.1 Instructor Overview

Chapter 12 is the horizon chapter — it looks at technologies that are entering educational AV now and will be mainstream within the course's useful lifespan. This is also the chapter most at risk of becoming outdated quickly, so the instructor's job is to teach evaluation frameworks, not just technology descriptions. Students should leave this chapter able to assess any new AV technology against a consistent set of criteria: What problem does it solve? For whom? At what cost and risk? How does it integrate with existing systems? What are the equity and accessibility implications? Key topics: AI-powered AV (intelligent cameras, speech-to-text, noise cancellation, automated content capture); Extended Reality (XR/VR/AR) in education; Internet of Things (IoT) sensors in building and AV management; LED video wall evolution; and the convergence of AV and building automation systems. The capstone presentations scheduled in Week 14 allow students to demonstrate synthesis learning — their AV Upgrade Plan should reference at least one emerging technology and evaluate it rigorously against the course's principles-first, equity-centered framework.

12.2 Lesson Plans

Session A — AI in Educational AV and Extended Reality

Time	Activity	Details
0–5 min	Warm-Up	Ask: "What AV technology do you expect will be completely different in 5 years? What do you think will remain the same? Why?"
5–25 min	Direct Instruction	AI in AV: intelligent camera tracking, AI noise cancellation (NVIDIA RTX Voice, Krisp), automatic speech recognition, automated lecture capture, AI-generated

Time	Activity	Details
		content summaries. Extended Reality: VR for immersive learning, AR for lab overlays and maintenance guidance, MR for collaborative design. Current educational deployments and evidence of effectiveness.
25–40 min	Active Learning	Technology evaluation exercise. Groups receive a hypothetical "emerging AV technology" pitch and must evaluate it against the course's criteria (problem solved, equity implications, cost, integration, evidence base). Present findings.
40–50 min	Debrief / Exit Ticket	Exit ticket: "Apply the technology evaluation criteria from today to an AI camera tracking system. What would you need to know before recommending it for a K–12 classroom?"

Session B — IoT, Building Convergence, and Capstone Presentations

Time	Activity	Details
0–5 min	Warm-Up	Brief review of capstone presentations scheduled for this session. Announce order. Remind audience of constructive feedback norms.
5–20 min	Direct Instruction	IoT in AV: occupancy sensors, environmental monitoring, room booking integration. Building automation convergence: BMS, AV, and IT integration. LED video

Time	Activity	Details
		wall evolution: fine-pitch LED, micro-LED, transparent LED. Sustainability in emerging AV: energy consumption, e-waste, longevity.
20–45 min	Active Learning	Capstone Presentations (Lab 12): Students present their AV Upgrade Plans (10 minutes each, with Q&A). Audience uses peer feedback form.
45–50 min	Debrief / Exit Ticket	Course reflection: "Name one principle from this course that you will apply in your professional work immediately. What will you do differently?"

12.3 Discussion Questions

61. AI systems in educational AV — including camera tracking and ASR captioning — are trained on data sets that may not adequately represent the diversity of voices, faces, and accents present in real classrooms. What are the equity implications and who is responsible for addressing them?
62. Virtual and augmented reality have been promoted as transformative educational tools for over a decade but have seen limited sustained adoption in K–12. What barriers (technical, pedagogical, cost, equity) explain this pattern, and what would need to change for XR to achieve broad adoption?
63. As AV systems become more integrated with building management systems (HVAC, lighting, occupancy, security), who owns the data generated by these integrated systems? What privacy protections should govern its use in educational settings?
64. LED video walls are increasingly replacing projection in large educational venues. From a total cost of ownership, image quality, and maintenance perspective, when is this substitution justified and when is it not?

65. The principles-first, vendor-agnostic, equity-centered philosophy of this course: how well do these principles equip you to evaluate emerging AV technologies that didn't exist when the course was designed? What additional frameworks or information sources would you use?

12.4 Differentiation Strategies

Learner Profile	Strategy
Beginners	Focus the technology evaluation exercise on one technology they have personally experienced (video conferencing AI, captioning). Familiar ground reduces cognitive load while teaching the evaluation framework.
Experienced teachers	Ask them to evaluate an emerging technology specifically from a pedagogical impact perspective: does it enhance or substitute for effective teaching? How does it change the instructor's role?
AV specialists	Assign current industry white papers on AV-over-IP at scale, micro-LED manufacturing advances, or LiDAR-based room sensing. Ask for a 5-minute "technology horizon" presentation to the class.

12.5 Lab Facilitation Notes

Capstone presentations require advance preparation: schedule presentation slots in Week 13, confirm technical requirements for each presenter (slide format, video clips, demo hardware), and test the room AV for presentation playback. Allow 10 minutes per presenter plus 3 minutes Q&A for a typical cohort of 12–15 students. Distribute peer feedback forms before presentations begin. Collect all feedback forms at the end and return to presenters after grading.

12.6 Common Student Misconceptions

Misconception	Correction Strategy
"AI will replace AV technicians and designers."	Discuss the persistent need for human judgment in AV design (needs assessment, stakeholder communication, ethical evaluation) and maintenance (physical diagnosis, context-sensitive troubleshooting). AI automates specific tasks, not professional judgment.
"Newer technology is always more sustainable."	Compare the embodied carbon, energy consumption, and disposal challenges of LED video walls vs. projectors. Newer does not automatically mean greener — lifecycle analysis is required.
"XR and VR are mature enough for everyday classroom use."	Present current limitations: headset cost, content development time, motion sickness rates, hygienic sharing concerns, and the limited peer-reviewed evidence of learning outcomes improvement. VR is powerful for specific, well-defined applications — it is not a general classroom tool yet.

12.7 Assessment Alignment Table

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
Q1-Q2	AI applications in educational AV	Domain 1, 7	2.1, 2.7	Analyze
Q3	XR/VR/AR in education	Domain 1	2.1, 2.5	Evaluate
Q4	IoT and AV/BMS convergence	Domain 1, 5	2.5	Analyze
Q5	LED video wall technology evolution	Domain 3	2.5	Analyze
Q6	Technology evaluation framework	Domain 1	2.7	Evaluate
Q7	Equity implications of	Domain 1, 11	2.3	Evaluate

Q#	Topic	AVIXA Domain	ISTE Standard	Bloom's
	AI in AV			
Q8	Sustainability in AV technology	Domain 1	2.3	Evaluate
Q9	Privacy in data-generating AV systems	Domain 1, 5	2.3	Analyze
Q10	Future-proofing AV infrastructure	Domain 8	2.7	Create

12.8 Accommodation Notes

XR/VR demonstrations during this chapter should not require student participation for those who experience motion sickness, vestibular disorders, or anxiety triggered by immersive environments. Provide an equivalent observation and analysis alternative. Capstone presentations should accommodate students who are not able to present orally — a recorded video presentation or written equivalent is an acceptable alternative if coordinated with disability services in advance.

12.9 Chapter Lab Rubric — Lab 12: Capstone Presentation

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Technical Accuracy	All technical claims accurate; emerging technology evaluated against sound criteria; AV design principles correctly applied	Mostly accurate; 1-2 minor technical errors	Several technical errors; emerging technology described but not evaluated	Significant inaccuracies; technology described uncritically
Procedure	Presentation meets all capstone	All sections present; one underdeveloped	2 required sections missing or	More than 2 sections missing;

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	requirements; all sections present and complete	d	very thin	requirements not met
Documentation	Written plan complete; evidence-based; recommendations supported with data and cost estimates	Plan complete; limited cost data	Plan present but thin; no cost estimates	No written plan or insufficient to evaluate
Teamwork	All team members presented; contributions equitable; collaborative Q&A	All members contributed; one dominated	Presentation primarily one member	Solo presentation despite team assignment

Assessment Instruments

INSTRUCTOR ONLY — DO NOT DISTRIBUTE TO STUDENTS

All assessment instruments in this section, including answer keys and exam questions, are for instructor use only. Reproduction for student distribution is a violation of the ISTE × AVIXA curriculum license agreement.

Midterm Exam — Chapters 1-6

Total Questions: 40 multiple-choice | **Time Allowed:** 60–75 minutes | **Passing Score:** 70% (28/40)

#	Question	A	B	C	D	Answer	AVIXA Domain
1	Which organization awards the CTS credential?	ISTE	IEEE	AVIXA	ANSI	C	Domain 1
2	The ISTE Standards for Educators were most recently updated in:	2017	2021	2023	2024	D	Domain 1
3	Which of the following best describes the vendor-agnostic approach?	Recommending the lowest-cost product	Evaluating products against performance criteria independent of brand	Using only open-source technology	Standardizing on one manufacturer	B	Domain 1
4	The decibel is best described as:	An absolute unit of sound pressure	A logarithmic ratio comparing two values	A measure of frequency	A unit of electrical voltage	B	Domain 2
5	According to the inverse square law, if you double the distance from a sound source, the SPL	3 dB	6 dB	10 dB	20 dB	B	Domain 2

#	Question	A	B	C	D	Answer	AVIXA Domain
	decreases by approximately:						
6	Which microphone type requires phantom power?	Dynamic	Ribbon	Condenser	Piezoelectric	C	Domain 2
7	RT60 refers to:	The time for a signal to travel 60 feet	The time for sound to decay by 60 dB	A type of cable connector	A network protocol for AV	B	Domain 2
8	The ANSI/ASA S12.60 standard addresses:	Display image size	Cable termination standards	Classroom acoustics	Video resolution	C	Domain 2
9	Throw ratio of a projector is calculated as:	Image width ÷ distance	Distance ÷ image width	Image height ÷ distance	Distance ÷ image height	B	Domain 3
10	Which display technology uses organic compounds to produce light without a backlight?	LCD	LED	OLED	DLP	C	Domain 3

#	Question	A	B	C	D	Answer	AVIXA Domain
11	The ANSI/AVIXA 2M standard recommends the maximum viewing distance for display content as:	4 × screen height	6 × screen height	8 × screen height	10 × screen height	B	Domain 3
12	What is HDCP primarily designed to prevent?	Signal degradation over long cables	Unauthorized copying of digital content	Network congestion	RF interference	B	Domain 3, 4
13	Which connector carries both audio and video signals in a single cable?	XLR	BNC	HDMI	RCA (composite video)	C	Domain 4
14	A signal distribution amplifier (DA) is used to:	Route any input to any output	Convert analog signals to digital	Send one input signal to multiple outputs	Increase cable bandwidth	C	Domain 4
15	Balanced audio signals use which	RCA	3.5mm TRS	XLR	BNC	C	Domain 4

#	Question	A	B	C	D	Answer	AVIXA Domain
	connect or type for professional AV?						
16	A matrix switcher differs from a distribution amplifier because it can:	Boost signal strength	Convert between signal formats	Route any of multiple inputs to any of multiple outputs	Eliminate cable length limitations	C	Domain 4
17	Digital signals experience a "cliff effect," meaning they:	Gradually degrade with increasing noise	Fail completely rather than degrading gradually	Require less bandwidth than analog	Are immune to interference	B	Domain 4
18	Which OSI model layer is most directly associated with IP addressing?	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)	Layer 7 (Application)	C	Domain 5
19	A VLAN is used in AV networking primarily to:	Increase Wi-Fi signal strength	Segment AV traffic from general network traffic	Assign static IP addresses	Replace the need for a control system	B	Domain 5
20	Dante is a protocol primarily used	Video over IP	Control signals	Professional audio over IP	Building automation commands	C	Domain 5

#	Question	A	B	C	D	Answer	AVIXA Domain
	for transporting:						
21	QoS in an AV network context stands for:	Queue of Services	Quality of Signal	Quality of Service	Queue of Streams	C	Domain 5
22	Which IP address is a valid Class C private address?	10.0.0.1	172.16.0.1	192.168.1.100	127.0.0.1	C	Domain 5
23	A network-connected AV device using the manufacturer's default password represents a:	Performance issue	Security vulnerability	Bandwidth limitation	Compatibility problem	B	Domain 5
24	Which of the following is the "brain" of an AV control system?	Touch panel	Control processor	Matrix switcher	Signal extender	B	Domain 6
25	In control system logic, a "macro" is best describe	A type of control cable	A series of automated commands triggered	A physical button panel	A software codec	B	Domain 6

#	Question	A	B	C	D	Answer	AVIXA Domain
	d as:		d by a single action				
26	RS-232 is a control communication protocol used primarily for:	Audio over IP	High-speed data transfer	Serial device control	Video distribution	C	Domain 6
27	A "distributed control architecture" means that control processing is:	Concentrated in a single central processor	Spread across multiple processors at device level	Managed entirely in the cloud	Eliminated in favor of direct input	B	Domain 6
28	Which ISTE Educator Standard is most closely aligned with data-informed decision making about technology use?	2.1 Learner	2.3 Citizen	2.5 Designer	2.7 Analyst	D	Domain 1
29	NDI is a protocol developed for:	Audio-over-IP	Video-over-IP	Control systems	Building automation	B	Domain 5
30	The primary	Replace the	Enable faster	Facilitate	Meet HDCP	C	Domain 4

#	Question	A	B	C	D	Answer	AVIXA Domain
	purpose of signal flow documentation is to:	need for signal processing	equipment purchasing	troubleshooting and system understanding	requirements		
31	A balanced audio cable achieves noise rejection through:	Using shielded wire only	Common-mode rejection of induced noise	Doubling the signal voltage	Reducing cable impedance	B	Domain 2, 4
32	Which standard governs AV systems performance verification?	ANSI/INFOCOMM 1M:2009	ANSI/AVIXA 2M:2010	IEC 60118-4	IEEE 802.1Q	A	Domain 8
33	When a teacher presses "Start Class" and the projector turns on but audio does not play, the FIRST troubleshooting step is to:	Replace the amplifier	Reboot the control processor	Define the problem and gather information	Call the AV vendor	C	Domain 6, 10
34	In AV over IP, multicast traffic	Improve QoS performance	Saturate network bandwidth	Eliminate the need for VLANs	Increase security	B	Domain 5

#	Question	A	B	C	D	Answer	AVIXA Domain
	that is not properly managed can:						
35	The abbreviation ANSI stands for:	Audio Network Systems Institute	American National Standards Institute	Association for Networked Signal Infrastructure	Applied Network Signal Integration	B	Domain 1
36	What is the primary advantage of static IP assignment for AV devices?	Reduces network security risk	Lower cost than DHCP	Ensures device address does not change	Increases bandwidth available to device	C	Domain 5
37	A condenser microphone is preferred over a dynamic microphone when:	High SPL environments require durability	Extended frequency response and sensitivity are required	Phantom power is unavailable	Used outdoors in rain	B	Domain 2
38	DisplayPort differs from HDMI in that it:	Cannot carry audio signals	Is proprietary to one manufacturer	Uses a royalty-free, open standard	Has a lower maximum bandwidth	C	Domain 4
39	The AV programming phase of a design project refers to:	Control system macro coding	Software installation on AV devices	Needs assessment and requirements gathering	Rack equipment layout	C	Domain 8

#	Question	A	B	C	D	Answer	AVIXA Domain
40	Which of the following is the best example of an equity-centered AV design decision?	Selecting a premium display brand for the boardroom	Installing hearing loop systems in all classrooms, not only auditoriums	Using wireless microphones for all instructors	Upgrading projectors in high-enrollment courses first	B	Domain 11

Final Exam — Chapters 7-12 + Cumulative

Total Questions: 50 multiple-choice | **Time Allowed:** 75-90 minutes | **Passing Score:** 70% (35/50)

Questions 1-35 cover Chapters 7-12; Questions 36-50 are cumulative (Chapters 1-12).

#	Question	A	B	C	D	Answer	AVIXA Domain
1	Which of the following best describes a UC&C "room system" as opposed to a software-only conferencing solution?	A system that requires no network connection	Dedicated hardware including a codec, camera, and microphone array designed for a specific room	A system that only supports one conferencing platform	A system managed by the IT department	B	Domain 7
2	In a	Single	Ceiling-	Multiple	Rear-of-	C	Domain

#	Question	A	B	C	D	Answer	AVIXA Domain
	hybrid classroom, which camera placement strategy best serves remote participant equity?	front-of-room fixed camera showing the instructor only	mounted 360-degree camera with AI tracking	cameras covering both instructor and student areas with switching capability	room fixed camera showing all students to the instructor		7
3	Mersive Solstice is an example of which technology category?	Digital signage CMS	Wireless content sharing/presentation	Video conferencing codec	AV-over-IP protocol	B	Domain 7
4	Beamforming microphone arrays in conference rooms work by:	Physically moving the microphone toward the speaker	Using multiple capsules and signal processing to focus audio pickup toward active speakers	Eliminating all ambient noise through cancellation	Routing audio over the network without amplification	B	Domain 7
5	The primary limitation of AI camera tracking in classrooms is:	High cost	Incompatibility with UC&C platforms	Failure in complex scenes with multiple people or rapid movement	Inability to capture 4K video	C	Domain 7

#	Question	A	B	C	D	Answer	AVIXA Domain
6	A "hybrid equity" concern in classroom AV design refers to:	Ensuring the room budget is divided fairly	Providing equivalent learning experiences for both in-room and remote participants	Using the same equipment brand throughout a campus	Ensuring all students use the same devices	B	Domain 7
7	The ANSI/AVIXA 2M standard establishes the minimum viewing distance as:	0.5 × screen height	1.0 × screen height	1.5 × screen height	2.0 × screen height	C	Domain 8
8	Which phase of the AV design process involves creating equipment lists and signal flow diagrams?	Programming	Schema Design	Design Development	Commissioning	C	Domain 8
9	A distributed speaker system in a classroom provides	Higher maximum SPL	More even audio coverage across all seating positions	Lower installation cost	Better bass frequency response	B	Domain 8

#	Question	A	B	C	D	Answer	AVIXA Domain
	which primary advantage over a single point-source speaker?						
10	During the AV programming phase, the most important stakeholders to consult are:	The AV manufacturer and the electrician	The facilities director and the IT manager	The end users who will operate and learn in the space	The building architect and the general contractor	C	Domain 8
11	A digital signage "CMS" stands for:	Cable Management System	Content Management System	Central Monitoring Station	Codec Media Server	B	Domain 9
12	The recommended dwell time for a single content zone on a digital sign is approximately:	2-3 seconds	5-7 seconds	8-12 seconds	20-30 seconds	C	Domain 9
13	Which digital signage configuration stores content locally	Cloud-based CMS with live streaming	SoC (System-on-Chip) display	Store-and-forward (on-premise player)	IPTV distribution system	C	Domain 9

#	Question	A	B	C	D	Answer	AVIXA Domain
	on the media player?						
14	ADA requires assistive listening systems in assembly areas that seat how many people?	25 or more	50 or more	100 or more	250 or more	B	Domain 9, 11
15	Minimum font size for digital signage content viewed at 10 feet is generally recommended to be at least:	12 points	24 points	36 points	48 points	C	Domain 9
16	Emergency alert system (EAS) integration with digital signage is most critical because it:	Reduces content production costs	Enables mass notification to replace printed emergency plans	Allows marketing to control safety messaging	Satisfies copyright requirements	B	Domain 9
17	The first step in the AV	Replace the most	Reboot the control	Define and document	Call the AV integrator	C	Domain 10

#	Question	A	B	C	D	Answer	AVIXA Domain
	troubles hooting methodology is to:	likely failed component	system	nt the problem	or		
18	Preventive maintenance of a projector most critically includes :	Replacing the lens annually	Cleaning or replacing the air filter per manufacturer schedule	Upgrading the firmware every quarter	Recalibrating the throw ratio monthly	B	Domain 10
19	Which document is most useful when troubleshooting an AV system that was installed by a different contract or 3 years ago?	The original equipment proposal	The manufacturer's marketing brochure	As-built signal flow diagrams and equipment schedule	The control system user manual	C	Domain 10
20	An AV system should be escalated to a certified technician (rather than addressed in-house) when:	Any component makes an unusual sound	The fault requires work inside energized equipment, structural mounting changes , or	The problem occurred more than once	The control panel shows an error message	B	Domain 10

#	Question	A	B	C	D	Answer	AVIXA Domain
			specialized test equipment				
21	Reactive maintenance differs from preventive maintenance in that it:	Costs less per incident	Occurs in response to failure rather than before failure	Is scheduled on a regular calendar basis	Is performed by the equipment manufacturer	B	Domain 10
22	ADA Title II applies to:	Private businesses only	Federal government facilities only	State and local government entities, including public schools	Nonprofit organizations only	C	Domain 11
23	An induction hearing loop system transfers audio to hearing aid users via:	Infrared light	FM radio frequency	Electromagnetic field	Bluetooth	C	Domain 11
24	WCAG 2.1 Level AA requires a minimum contrast ratio for normal body	2.1:1	3.0:1	4.5:1	7.0:1	C	Domain 11

#	Question	A	B	C	D	Answer	AVIXA Domain
	text of:						
25	Universal Design for Learning (UDL) in AV design means:	Designing one version of the system for average users	Designing systems that are flexible and accessible for the widest range of users from the outset	Providing a separate accessible version upon request	Limiting features to reduce complexity for all users	B	Domain 11
26	The social model of disability, applied to AV, implies that:	Disability is a permanent characteristic of an individual	People should adapt to AV systems as designed	AV systems that exclude users create the disabling barrier, not the person	Accessibility is optional if the majority of users are unaffected	C	Domain 11
27	Section 508 of the Rehabilitation Act applies to AV systems that are:	Used in any US school district	Procured or developed by federal agencies	Sold by AVIXA member companies	Installed in buildings over 10 years old	B	Domain 11
28	Audio description in AV accessibility provides:	A text transcript of spoken dialogue	A verbal narration of visual information for users who are Blind or have	Amplified audio for users who are hard of hearing	A simplified version of complex content	B	Domain 11

#	Question	A	B	C	D	Answer	AVIXA Domain
			low vision				
29	Which of the following AI tools is specifically designed to reduce background noise in video conferencing audio?	AI camera tracking	Automatic speech recognition (ASR)	AI noise cancellation (e.g., NVIDIA RTX Voice)	AI-generated content summarization	C	Domain 7, 12
30	In the context of emerging AV, XR is an umbrella term that includes :	Xero-copy reproduction	VR, AR, and MR	Extended range networking	XML-based room configuration	B	Domain 12
31	A key limitation of VR in K-12 educational settings is:	Inability to display color	Cost, content development time, and equity of access per headset	Incompatibility with school networks	Inability to display 3D content	B	Domain 12
32	IoT sensors in AV-enabled educational	Automated control of HVAC, lighting,	Direct student assessment data	Replacement for control system processors	Video streaming to remote sites	A	Domain 12

#	Question	A	B	C	D	Answer	AVIXA Domain
	spaces can provide:	and AV based on occupancy					
33	Micro-LED technology differs from conventional LED displays primarily in:	Using smaller LEDs enabling higher resolution and brightness at smaller pixel pitches	Using liquid crystal backlit by LEDs	Requiring less power at all screen sizes	Being exclusively manufactured for outdoor use	A	Domain 12
34	When evaluating an emerging AV technology for educational adoption, which consideration is MOST aligned with the equity-centered philosophy of this course?	Whether it was demonstrated at a trade show	Who within the school will be able to use it independently, and who will be excluded	Whether a peer district has already adopted it	Whether it comes with a warranty	B	Domain 12
35	Future-proofing an AV infrastructure most effectively	Purchasing the highest-specification equipment available	Designing for flexible signal pathways, structured	Installing equipment that will not need replacement for	Using only wireless systems to avoid cable obsolescence	B	Domain 8, 12

#	Question	A	B	C	D	Answer	AVIXA Domain
	involves :	e today	cabling, and upgrade able processing	20 years			
36	Which of the following best describes the function of a scaler in a signal chain?	Amplifies signal strength over long distances	Converts a signal from one resolution or format to another	Splits one signal into multiple outputs	Encrypts digital content	B	Domain 4
37	A properly installed hearing loop system in a classroom primarily benefits students who:	Have visual impairments	Wear hearing aids or cochlear implants with telecoil (T-coil) capability	Have cognitive processing differences	Are non-native English speakers	B	Domain 2, 11
38	The principle s-first approach in this course means instructors should prioritize teaching :	Specific product configuration procedures	The most recent manufacturer certification requirements	Underlying concepts that remain valid across evolving technologies	How to pass the CTS exam in the shortest time	C	Domain 1
39	According to the	45 dB	80 dB	84 dB	87 dB	C	Domain 2

#	Question	A	B	C	D	Answer	AVIXA Domain
	inverse square law, if the distance from a speaker doubles, SPL decreases by approximately 6 dB. If the original SPL at 5 feet is 90 dB, what is the approximate SPL at 10 feet?						
40	Which ISTE Educator Standard is most aligned with designing technology-enhanced learning experiences?	2.1 Learner	2.3 Citizen	2.5 Designer	2.7 Analyst	C	Domain 1
41	A classroom's existing project	2 years	3 years	4 years	5 years	C	Domain 10

#	Question	A	B	C	D	Answer	AVIXA Domain
	r has a 6,000-hour lamp life. The projector is used 8 hours/day, 180 days/year. After how many years should the lamp be replaced?						
42	A networked AV system's control process or is unresponsive. After confirming it has power, what is the NEXT recommended troubleshooting step?	Replace the control process or immediately	Check network connectivity and verify the processor's IP address is reachable	Reprogram all macros from scratch	Contact the manufacturer for a replacement	B	Domain 6, 10
43	Which of the following is NOT a component of a	Media player	Content management system	Display	Matrix switcher	D	Domain 9

#	Question	A	B	C	D	Answer	AVIXA Domain
	digital signage system?						
44	In a hybrid classroom, which improvement most directly benefits the remote participant's audio experience?	Upgrading the in-room display to 4K	Adding a second monitor for the instructor	Installing a ceiling microphone array with acoustic echo cancellation	Upgrading the control system touch panel	C	Domain 7
45	Which type of cable is MOST appropriate for carrying a balanced professional audio signal between two rack-mounted devices over 20 feet?	RG-6 coaxial	CAT6 Ethernet	2-conductor or shielded with XLR connectors	HDMI 2.1 cable	C	Domain 2, 4
46	The primary purpose of commissioning	Demonstrate the system to the client	Verify that the system performs to the specifications	Train all end users on system operation	Transfer ownership to the facilities department	B	Domain 8, 10

#	Question	A	B	C	D	Answer	AVIXA Domain
	an AV system after installation is to:	for marketing	d design standards				
47	Which accessibility feature directly addresses the needs of a student who is Blind during a video-based lecture?	Captioning	Hearing loop	Audio description	High-contrast display	C	Domain 11
48	AVB/TSN (IEEE 802.1) is distinguished from proprietary AV-over-IP protocols like Dante primarily by being:	Limited to audio transport only	An open, standards-based network transport specification	Available only for large enterprise systems	Requiring specialized network hardware unavailable from standard IT vendors	B	Domain 5
49	A school administrator asks you to justify the investment in an AV	It is required by the AV warranty	It keeps the equipment looking clean	It reduces total cost of ownership by preventing expensive	It allows the IT department to avoid AV support calls	C	Domain 10

#	Question	A	B	C	D	Answer	AVIXA Domain
	preventive maintenance program. The STRONGEST argument is:			reactive failures and system downtime			
50	Which combination of AV principles from this course best defines a professional, ethical approach to recommending an AV system for a school?	Specifying the newest technology at the highest available price point	Recommending the vendor with whom you have the longest relationship	Evaluating options against performance criteria, accessibility requirements, and total cost of ownership without vendor bias	Standardizing on whatever the neighboring district uses	C	Domain 1, 11

Lab Rubric Master Template (Reusable)

This rubric is designed for reuse across all 12 chapter labs. Fill in the Lab Name and adjust criterion descriptors as specified in each chapter's section 9 rubric. The 4-point scale is consistent throughout the course.

Criterion	4 — Exemplary (100%)	3 — Proficient (80%)	2 — Developing (65%)	1 — Beginning (50%)	Weight
Technical Accuracy <i>Correctness of AV concepts, calculations, and identifications</i>	All technical content is correct; terminology and concepts applied with precision; no factual errors	Mostly correct; 1-2 minor errors that do not affect overall understanding	Several errors; demonstrates partial understanding of core concepts	Significant errors throughout; core concepts not demonstrated	30%
Procedure <i>Completion of all required lab steps</i>	All required steps completed; procedures followed correctly and in sequence	All steps attempted; minor procedural deviation that did not affect outcome	Most steps completed; significant deviation from procedure or missing steps	Fewer than half the steps completed; procedure not followed	25%
Documentation <i>Quality of written lab report, diagrams, and records</i>	Documentation complete, organized, and professionally written; all required components present	Documentation mostly complete; minor omissions; generally clear	Documentation incomplete; significant omissions; some content unclear	Documentation absent or insufficient; no usable information recorded	25%
Teamwork <i>Quality of collaboration and equitable participation</i>	All team members contributed equitably; evidence of supportive communication and shared problem-solving	Generally equitable; one member slightly more dominant	One or two members dominated; others minimally engaged	No evidence of collaboration; individual work submitted as group	20%

Score Conversion: Multiply each criterion score (1-4) by its weight, sum the weighted scores, and convert: 4.0 = 100%, 3.0 = 80%, 2.0 = 65%, 1.0 = 50%. Instructors may apply the formula: $(\text{sum of weighted criterion scores} \div 4) \times 100 = \text{percentage grade}$.

Participation and Professionalism Rubric

Assessed weekly by instructor observation. Counts toward the 10%

Participation/Professionalism course grade component.

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Attendance and Punctuality	Present and on time for all sessions; absences rare and communicated in advance	Occasional tardiness or one unexcused absence	Multiple instances of tardiness or 2 unexcused absences	Frequent absences or tardiness that disrupts learning
Engagement and Contribution	Consistently contributes relevant, substantive responses; asks questions that advance class understanding	Regularly participates; contributions mostly relevant	Participates when prompted; limited voluntary contributions	Rarely participates; off-task or disruptive behavior
Respect and Inclusivity	Actively inclusive of all perspectives; models respectful discourse; does not dominate or exclude	Generally respectful; one or two instances of interruption or dismissiveness	Occasional behavior that limits others' participation	Repeated disrespectful behavior; creates exclusionary environment
Professional Conduct	Meets all deadlines; communicates proactively about challenges; treats course materials and equipment with care	Generally professional; one lapse in deadline or communication	Multiple deadline misses; limited proactive communication	Consistent failure to meet professional expectations

Course Portfolio Rubric

The course portfolio is due at the end of Week 14. It compiles evidence of learning across the full course. Students submit digitally via the LMS.

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Completeness	All required artifacts included: lab reports (12), exit tickets (select 6), midterm and final reflections, capstone	10-11 lab reports; 5-6 exit tickets; midterm or final reflection	7-9 lab reports; 3-4 exit tickets	Fewer than 7 lab reports; minimal portfolio content
Reflection Quality	Reflections demonstrate genuine engagement with course concepts; evidence of changed thinking or practice cited specifically	Reflections present; some specificity; generally thoughtful	Reflections superficial or summary-level; limited evidence of metacognition	No meaningful reflection; artifacts listed without commentary
Connections to ISTE and AVIXA Standards	Portfolio introduction explicitly maps artifacts to ISTE Educator Standards and AVIXA domains; connections are specific and accurately cited	Standards connections present; some mappings vague	Standards mentioned but not meaningfully connected to specific artifacts	No standards connections
Organization and Presentation	Portfolio is logically organized with a table of contents, consistent formatting,	Organized; minor formatting inconsistencies	Some organizational logic; difficult to navigate	Disorganized; no table of contents; inaccessible formatting

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
	and accessible file formats			

Capstone Project: Design an AV Upgrade Plan for Your School or Classroom

Project Description

The capstone project is a professional-quality AV upgrade plan for a real educational space — ideally the student's own classroom, department, or institution. Students who do not have a real-world site may use a provided hypothetical case study. The plan applies all major course concepts to a genuine, equity-centered design problem. It is submitted as a written document and presented to the class in Week 14.

Required Components

66. **Site Analysis (20%):** Description of the existing space; current AV inventory; documented deficiencies against applicable ANSI/AVIXA standards; accessibility audit findings; user needs assessment (must include input from at least 2 end users)
67. **Learning Objectives and User Requirements (10%):** Clear statement of what the upgraded system must enable educationally; documented requirements from stakeholder consultation
68. **System Design (30%):** Recommended display system with specifications and sight-line justification; audio system with coverage analysis; signal flow diagram; network requirements (IP plan, VLAN, QoS); control system approach with at least one macro flowchart
69. **Accessibility Plan (15%):** How the design addresses ADA, Section 508 (if applicable), and UDL principles; specific accessibility features included and justification

70. **Budget and Implementation Plan (15%):** Line-item budget estimate (vendor-agnostic categories and approximate cost ranges); proposed implementation phases; timeline

71. **Emerging Technology Evaluation (10%):** Identification of one emerging technology relevant to the site; evaluation against the course criteria (problem solved, equity implications, evidence, integration, cost)

Sample Exemplar Outline

Section	Exemplar Content Summary	Target Length
Executive Summary	One-page summary of key findings and recommendations; cost summary; top accessibility improvement	1 page
Site Analysis	Floor plan with dimensions; photos of current equipment; current vs. standard comparison table; 3-person user interview summary	3-4 pages
User Requirements	Prioritized requirement list from interviews; educational use cases (lecture, group work, hybrid, assessment)	1-2 pages
System Design	Display: 86" LED panel at 12-ft viewing distance (meets ANSI/AVIXA 2M); Audio: distributed ceiling array with echo cancellation; Signal: HDMI 2.0 matrix + USB-C at each table; Network: dedicated VLAN, static IPs, QoS for Dante; Control: single touch panel with 4 preset macros (flowcharts included)	4-6 pages
Accessibility Plan	Hearing loop installed	1-2 pages

Section	Exemplar Content Summary	Target Length
	(IEC 60118-4); all displays at ADA-compliant height; captioning enabled via Zoom integration; tactile labels on control panel	
Budget and Implementation	Total estimated cost: \$28,000-\$35,000; Phase 1 (critical): display + audio (\$18K); Phase 2 (Year 2): control system + loop (\$15K); contingency: 15%	1-2 pages
Emerging Technology	AI camera tracking evaluated: benefits (engagement, remote equity), risks (privacy, bias in tracking dark-skinned faces), evidence base (limited peer-reviewed studies), integration (works with existing Zoom Room), cost (\$3,200 camera system); Recommendation: pilot in one room before campus-wide adoption	1-2 pages

Capstone Project Rubric

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning	Weight
Site Analysis and Needs Assessment	Comprehensive analysis with measurements, photos, standards comparison, and documented user input from ≥ 2 stakeholder	Solid analysis; one element (user input, standards comparison, or physical data) underdeveloped	Basic analysis; site data incomplete; user input absent or anecdotal	Minimal or no site analysis; no user input	20%

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning	Weight
	s				
System Design Quality	All four system components specified (display, audio, signal, control); all design decisions justified by standards, calculations, or requirements; signal flow diagram complete	All components addressed; 1-2 decisions lack justification	2-3 components addressed; design decisions not consistently justified	Fewer than 2 components; no justification	30%
Accessibility Integration	ADA, UDL, and relevant standards applied throughout design (not as a separate add-on); specific features cited with standard references	Accessibility addressed in a dedicated section; features cited; not fully integrated into design rationale	Accessibility mentioned but features vague or compliance not verified	Accessibility absent or treated as optional	20%
Budget and Feasibility	Line-item budget with realistic cost ranges; phased implementation plan; cost justified against educational	Budget present with general estimates; one phase defined	Budget too vague to evaluate; no implementation phases	No budget or placeholder numbers only	15%

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning	Weight
	outcome				
Emerging Technology Evaluation	Technology evaluated rigorously against all course criteria; equity implications addressed; clear, evidence-based recommendation	Evaluation present; one criterion missing or shallow	Technology described but not critically evaluated	Emerging technology absent or included as marketing summary only	15%

Recommended Grading Structure

The following grading weights are recommended for the 14-week configuration. Instructors may adjust weights within $\pm 5\%$ to meet institutional grading policy requirements, provided the adjustments are documented in the syllabus and applied consistently.

Component	Weight	Details	Dropped / Flexibility
Participation and Professionalism	10%	Assessed weekly using the Participation and Professionalism Rubric; averaged across 14 weeks	Lowest 2 weeks dropped to account for illness or emergencies
Weekly Lab Reports (12 labs)	30%	Each lab report graded using the Lab Rubric Master Template (4-point, 4-criterion); all 12 labs weighted equally at 2.5%	Lowest 1 lab score dropped; late labs accepted within 1 week at 10% penalty

Component	Weight	Details	Dropped / Flexibility
		each	
Midterm Exam	20%	40 multiple-choice items (Chs. 1-6); 60-75 minutes; machine- or hand-scored	One retake opportunity offered within 1 week for students scoring below 60%; retake score averaged with original
Chapter Quizzes (best 8 of 12)	15%	10-question quiz at end of each chapter; drawn from textbook review questions; auto-graded in LMS	Best 8 of 12 counted; lowest 4 automatically dropped, no documentation required
Final Exam	15%	50 multiple-choice items (Chs. 7-12 + cumulative); 75-90 minutes; taken during Week 14 exam period	No retake; extended time accommodations apply per disability services documentation
Capstone Project	10%	Written AV Upgrade Plan + 10-minute class presentation; graded using the Capstone Rubric; submitted Week 14	Up to 1-week extension available with written request submitted by Week 13; no further extensions
TOTAL	100%		

Grade Scale

Percentage	Letter Grade	Note
93-100%	A	Exemplary mastery of all course outcomes
90-92%	A–	
87-89%	B+	
83-86%	B	Proficient; meets all professional readiness

Percentage	Letter Grade	Note
		benchmarks
80–82%	B–	
77–79%	C+	
73–76%	C	Satisfactory; minimum for AVIXA AV Technologist eligibility recommendation
70–72%	C–	
60–69%	D	Below expectations; does not meet professional readiness benchmarks
Below 60%	F	Course not passed; recommend retake

Instructor Tip — Incomplete Grade Policy

Establish your institution's Incomplete (I) grade criteria clearly in the syllabus.

Students who have completed at least 70% of coursework with passing marks and face documented extenuating circumstances are generally eligible for an I grade.

Set a clear completion deadline — typically the following semester — and document the agreement in writing.

Chapter Review Question Answer Keys

INSTRUCTOR ONLY — Answer Keys

These answer keys cover all 120 chapter review questions (10 per chapter × 12 chapters). Each answer includes the correct response, a 2–3 sentence explanation, the primary AVIXA domain, and the aligned ISTE Educator Standard. Do not distribute to students.

Chapter 1 Answer Key: The AV Ecosystem

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	B — Audiovisual systems encompass electronic technologies for capturing, processing, transmitting, and displaying audio and visual content	AV is a distinct discipline from consumer electronics or IT, focused on the design, integration, and performance of systems that manage sound and image in professional and educational contexts. The definition excludes purely IT-managed data systems without audio or visual output.	Domain 1	2.1 Learner
2	C — AVIXA is the trade association and certification body for the AV industry, publishing	AVIXA (formerly InfoComm) serves as the professional home for AV practitioners worldwide,	Domain 1	2.1 Learner

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	standards and awarding CTS credentials	setting training and certification standards through the CTS program. It is distinct from ISTE, which focuses specifically on educational technology for teaching and learning.		
3	A — The ISTE Educator Standards describe competencies for teachers in using technology effectively and equitably	The 2024 v4.02 ISTE Educator Standards establish seven competency areas (Learner, Leader, Citizen, Collaborator, Designer, Facilitator, Analyst) for educators. They are not certification requirements but a professional development framework adopted across all U.S. states.	Domain 1	2.1, 2.3
4	D — AV design addresses acoustics, room geometry, signal integrity,	While AV and IT increasingly converge through networked systems, AV design requires	Domain 1	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	and human factors not covered by standard IT curricula	physical principles (acoustics, optics) and human factors knowledge (sight lines, hearing coverage) that are outside the scope of IT networking certifications. This distinction justifies AV as a separate professional discipline.		
5	B — AV Technologist (entry) → CTS → CTS-D (design) or CTS-I (installation)	AVIXA structures its credential pathway to progressively validate increasing levels of AV competency. The AVIXA-Recognized AV Technologist Certificate is a non-accredited entry point; the CTS is the primary ANAB-accredited professional credential; CTS-D and CTS-I are specialty credentials requiring a valid CTS.	Domain 1	2.1 Learner
6	C — ANSI, IEC, and ISO are	Standards bodies (ANSI, IEC, ISO)	Domain 1	2.3 Citizen

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	standards bodies; AVIXA and ISTE are professional associations	develop technical specifications that define performance thresholds and testing methodologies . Professional associations like AVIXA and ISTE develop training, certification, and advocacy programs. The two types of organizations serve complementary but distinct roles.		
7	A — Voluntary standards set minimum performance thresholds; ADA and local building codes are legally enforceable	Most AVIXA-published standards (e.g., ANSI/AVIXA 2M) are voluntary consensus standards — professionally authoritative but not legally mandated in most jurisdictions. ADA requirements and local electrical codes carry legal force. Designers must understand which standards are enforceable	Domain 1	2.3 Citizen

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		and which are best-practice benchmarks.		
8	B — Poor AV infrastructure (unintelligible audio, inaccessible displays) creates barriers that fall disproportionately on students with disabilities and in under-resourced schools	Equity in educational AV is not merely an ethical concern — it is a measurable outcome driver. Research links room acoustics to reading achievement, display quality to attention, and hearing loop availability to participation by students who are hard of hearing. AV design decisions are education equity decisions.	Domain 1	2.3 Citizen
9	C — Vendor-agnostic evaluation means assessing products against defined performance criteria independent of brand relationships or marketing	Vendor-agnostic practice protects institutional stakeholders from biased recommendations and ensures that AV investments are made on technical and educational merit. It requires establishing	Domain 1	2.7 Analyst

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		evaluation criteria before examining products, not after.		
10	A — Complete the 100-question AVIXA-Recognized AV Technologist Test with a passing score of 80%	The AV Technologist Certificate requires passing a 90-minute, 100-question online test with a minimum score of 80%. It is free for AVIXA Premium and Elite members and serves as a bridge credential toward the CTS. No prior coursework is required, though the CTS 1: AV Technology course is recommended preparation.	Domain 1	2.1 Learner

Chapter 2 Answer Key: Audio Fundamentals

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	B — Frequency is the number of cycles per second (Hz); wavelength	Frequency and wavelength are inversely related: higher frequency = shorter	Domain 2	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	is the physical distance of one cycle	wavelength. At 20°C, the speed of sound is approximately 343 m/s; wavelength = speed ÷ frequency. A 1 kHz tone has a wavelength of approximately 34 cm, which has implications for speaker placement and room resonance.		
2	C — The decibel is a logarithmic ratio expressing the relationship between two power or pressure values	Because human hearing perceives loudness on a logarithmic scale, the decibel is a natural unit for acoustic measurement. A 10 dB increase corresponds roughly to a perceived doubling of loudness and represents a tenfold increase in acoustic power. This logarithmic nature means small dB numbers represent large real-world	Domain 2	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		differences.		
3	A — Using the inverse square law: SPL decreases by 6 dB each time distance doubles	The inverse square law states that intensity is inversely proportional to the square of the distance from the source. For every doubling of distance, SPL decreases by approximately 6 dB. This is foundational for speaker placement calculations and determining coverage uniformity across a room.	Domain 2	2.5 Designer
4	D — Dynamic microphones use electromagnetic induction; condenser microphones use a capacitive capsule; ribbon microphones use a thin metallic ribbon	Each transducer type has a distinct operating principle and appropriate use case. Dynamic mics are robust and suited for high-SPL environments; condensers offer superior sensitivity and extended frequency response but require phantom power; ribbons	Domain 2	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		offer a distinctive frequency response but are fragile.		
5	B — Balanced signals use two conductors carrying equal and opposite signals; noise induced on both conductors is cancelled at the receiver (common-mode rejection)	Balanced audio transmission is the professional standard for cable runs because it rejects electromagnetically induced noise that would otherwise add hum or interference. Balanced connections use XLR or TRS connectors; unbalanced connections use RCA or TS connectors and are susceptible to noise over long runs.	Domain 2, 4	2.5 Designer
6	C — Reflection (sound bouncing off surfaces), absorption (sound energy converted to heat), and diffusion (sound scattered in multiple	The balance of reflection, absorption, and diffusion in a room determines its acoustic character. A room with excessive reflection has long reverberation times,	Domain 2	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	directions) are the three primary acoustic phenomena	degrading speech intelligibility. A room with excessive absorption may sound acoustically "dead." Diffusion scatters energy without shortening reverberation, improving perceived spatial quality.		
7	A — RT60 is the time required for a sound to decay by 60 dB after the source stops	RT60 is the primary metric for characterizing room acoustics. ANSI/ASA S12.60 specifies RT60 targets for classrooms (≤ 0.6 seconds for core learning spaces ≤ 566 m ³). High RT60 values indicate excessive reverberation, which reduces speech intelligibility particularly at consonant frequencies.	Domain 2	2.5 Designer
8	B — ANSI/ASA S12.60 specifies	ANSI/ASA S12.60 is the American standard for	Domain 2	2.3 Citizen

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	that background noise in core learning spaces shall not exceed 35 dBA (NC-35)	classroom acoustics and directly informs AV system design. A classroom exceeding NC-35 requires the instructor to raise their voice significantly, causing vocal fatigue and reducing intelligibility for students who are hard of hearing. Many existing classrooms significantly exceed this standard.		
9	C — Sound field amplification systems amplify the instructor's voice to a consistent level throughout the room, reducing vocal strain and improving intelligibility for all students	Research shows that sound field amplification systems improve academic outcomes for all students, not only those with hearing differences. They are particularly beneficial in classrooms with elevated ambient noise levels. Teachers using sound field systems report significantly	Domain 2	2.5, 2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		reduced vocal fatigue over the academic year.		
10	A — Speaker coverage pattern (dispersion angle) must match the audience area geometry; a speaker with too narrow a pattern will create uneven coverage	Speaker selection requires matching the dispersion angle (specified as horizontal x vertical degrees at –6 dB points) to the width and depth of the audience area. A speaker covering a wide room requires wide horizontal dispersion; a speaker covering a deep, narrow room requires controlled horizontal and extended vertical dispersion. Mismatched patterns create "hot spots" and "dead zones."	Domain 2	2.5 Designer

Chapter 3 Answer Key: Video Fundamentals

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	B —	Resolution is	Domain 3	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	Resolution describes the number of pixels in an image; 1920×1080 (1080p) has approximately 2 megapixels	one factor in perceived image quality — viewing distance, luminance, and contrast ratio are equally important. At typical classroom viewing distances, the human visual system cannot resolve 4K detail on a standard-sized display, making 4K's educational benefit context-dependent rather than universal.		
2	D — Pixel density (PPI) determines perceived sharpness at a given viewing distance; higher PPI is only perceptually beneficial within a threshold viewing distance	The viewing distance formula ($D = H \times N$, where H is screen height and N is a multiplier based on resolution and visual acuity) determines whether increased resolution is perceptible to viewers. Beyond approximately 8H for 1080p content, human visual acuity limits	Domain 3	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		prevent the perception of additional detail from higher resolutions.		
3	A — LCD (liquid crystal display) requires a backlight; OLED (organic LED) generates light from organic compounds without a separate backlight; LED (in flat-panel context) typically refers to an LCD with LED backlight; laser projectors use laser light sources	Each display technology has distinct characteristics for educational applications: LCD/LED panels offer long life and bright images; OLED offers superior contrast but has burn-in risk and higher cost; laser projectors offer extended lamp life over traditional lamp projectors. Technology selection should follow a requirements-driven process.	Domain 3	2.5 Designer
4	C — Throw distance = throw ratio × image width; a 1.5:1 throw ratio projector producing a 100" wide image must be placed 150" (12.5 feet) from the screen	Throw ratio is the fundamental specification for projector placement. Standard throw ratios range from approximately 1.3:1 to 2.0:1; short-throw ratios are typically	Domain 3	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		0.4:1–0.8:1; ultra-short-throw (UST) projectors have ratios below 0.4:1. Misapplying throw ratio calculations results in incorrectly sized images or projectors that cannot be physically positioned.		
5	B — Ambient light reduces effective image contrast; a 5,000-lumen projector may appear dim in a bright room because ambient light raises the black level of the image	Display luminance specifications (ANSI lumens for projectors; nits for flat panels) must be evaluated relative to ambient light conditions. The ambient light rejection (ALR) capability of a screen and room light control measures are as important as projector lumens in determining perceived image quality in real-world classrooms.	Domain 3	2.5 Designer
6	A — The minimum viewing distance (near limit)	ANSI/AVIXA 2M:2010 establishes both minimum (1.5H) and	Domain 3	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	is 1.5 × screen height (1.5H) per ANSI/AVIXA 2M	maximum (6H for standard content, 4H for detailed work) viewing distances based on content type and display resolution. Seats within 1.5H may experience excessive neck strain or visible pixelation; seats beyond 6H may not be able to resolve fine detail.		
7	C — For a 60-foot deep room: if screen height is 10 feet, maximum viewing distance = $6 \times 10 = 60$ feet; the rear seats exactly meet the standard	Applying the 6H maximum viewing distance rule: screen height must be at least room depth ÷ 6. For a 60-foot room, the minimum screen height is 10 feet. Many older educational facilities have inadequate screen sizes for their room depths — this is a common finding in accessibility and AV audits.	Domain 3	2.5 Designer
8	B — sRGB is the standard color space	Color space selection affects color	Domain 3	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	for web and general computing; DCI-P3 is a wider color space used in digital cinema and professional displays	accuracy across the content production and display workflow. Educational settings typically use sRGB-calibrated displays for standard presentations; disciplines requiring wider color gamut (art, design, medical imaging) benefit from DCI-P3 capable displays. HDR (High Dynamic Range) extends luminance range beyond standard content.		
9	A — Sensor size affects low-light performance and depth of field; larger sensors capture more light and allow shallower depth of field	Camera sensor size is a primary determinant of image quality in variable lighting conditions, which are common in educational spaces. Frame rate (24fps for cinematic, 30fps for video, 60fps for smooth	Domain 3	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		motion) affects motion representation . Codec selection (H.264, H.265, AV1) determines compression efficiency and bandwidth requirements for streaming.		
10	D — Sight line analysis ensures all seats have an unobstructed view of the display that meets minimum resolution and angular criteria; poor sight lines are an equity issue affecting which students can fully access instructional content	Sight line analysis is the process of verifying that every seat in a room has an acceptable view of the primary display — free of obstructions, within the required distance range, and at an acceptable vertical and horizontal angle. Students in seats with poor sight lines receive unequal access to visual instructional content, making sight line design an equity issue.	Domain 3, 8	2.5 Designer

Chapter 4 Answer Key: Signal Flow

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	C — Analog signals are continuous electrical representations of physical phenomena; digital signals are discrete numerical representations encoded in binary	The distinction between analog and digital domains is foundational to understanding AV signal flow. Conversion between domains occurs at ADC (analog-to-digital converter) and DAC (digital-to-analog converter) stages. Each conversion introduces small amounts of error, so minimizing domain conversions is a design best practice.	Domain 4	2.5 Designer
2	B — Digital signals experience a "cliff effect" — they are error-free up to a degradation threshold, then fail completely rather than degrading gradually	This characteristic distinguishes digital signal behavior from analog: an analog signal becomes noisy progressively as quality degrades, while a digital signal remains perfect until the error correction	Domain 4	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		limit is exceeded, at which point it fails entirely. This makes digital systems generally more reliable but harder to diagnose from degradation symptoms alone.		
3	A — HDMI (video+audio), XLR (professional balanced audio), BNC (SDI video, RF), RJ-45 (network/Ethernet), DisplayPort (video+audio)	Connector identification is a fundamental AV professional competency. Each connector type is associated with specific signal types, voltage levels, and impedance characteristics. Using incorrect connectors or adapters without understanding signal compatibility is a common cause of AV system faults.	Domain 4	2.5 Designer
4	D — A block diagram shows signal sources, processing devices,	Block diagrams are the primary documentation tool for AV system signal	Domain 4	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	distribution paths, and destinations with directional signal flow arrows; it does not show physical cable routing or equipment dimensions	flow. They use standardized symbols and directional arrows to represent how signals travel through a system. Reading and creating block diagrams is prerequisite to troubleshooting and system expansion work.		
5	A — A distribution amplifier (DA) takes one input and sends it to multiple outputs simultaneously, buffering the signal to prevent loading	Distribution amplifiers prevent the signal degradation that occurs when a source is passively split to multiple displays. Each output of a DA is a fully buffered copy of the input. 1×4 DAs (one input, four outputs) are among the most common AV infrastructure devices in educational settings.	Domain 4	2.5 Designer
6	C — A matrix switcher routes any of its inputs to any of its outputs	Matrix switchers are used when flexible routing of multiple	Domain 4	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	independently; a 4×4 matrix can send input 1 to output 3, input 2 to output 1, and input 4 to output 2 simultaneously	sources to multiple destinations is required — a common need in lecture halls, divisible rooms, and campus distribution systems. They are more complex and costly than distribution amplifiers but provide routing flexibility that DAs cannot.		
7	B — HDCP (High-bandwidth Digital Content Protection) is a digital rights management protocol that can cause "handshake failures" when non-HDCP-compliant devices are in the signal path	HDCP was designed for consumer entertainment content protection but creates significant problems in professional AV environments where signal distribution, recording, and switching are routine. Designing HDCP-compliant signal chains (using compliant switchers, extenders, and displays) prevents these failures. Instructors	Domain 4	2.3 Citizen

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		should note that this is a design responsibility, not a user error.		
8	A — Impedance matching ensures maximum power transfer between source and load; mismatched impedance causes signal reflections, frequency response errors, and power loss	Impedance (measured in ohms) must be matched between source output and load input for optimal signal transfer. In professional audio, 600-ohm output driving a 10k-ohm input is a bridging configuration optimized for voltage transfer rather than power. Speaker impedance matching is particularly critical to prevent amplifier damage.	Domain 4	2.5 Designer
9	C — Eliminate → Isolate → Replace: first confirm the problem is real and defined; then systematically narrow the fault location; then replace	The troubleshooting methodology introduced in Chapter 4 and formalized in Chapter 10 is a transferable professional skill applicable beyond AV. Beginning with elimination (is the input	Domain 4, 10	2.7 Analyst

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	or repair the identified faulty component	source working? is the display powered?) before isolation (which segment of the chain has the fault?) prevents premature and costly component replacement.		
10	B — A signal flow diagram documents every component and connection in a system; it is the primary reference for troubleshooting, expansion, and handover to new personnel	Signal flow diagrams have professional value beyond installation: they enable any qualified technician to understand the system without prior familiarity, they document what is actually installed (as opposed to what was designed), and they provide the baseline for troubleshooting. Their absence is one of the most commonly cited barriers to effective AV support in educational institutions.	Domain 4	2.5 Designer

Chapter 5 Answer Key: Networking for AV

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	C — Layer 1 (Physical): cabling and connectors; Layer 2 (Data Link): MAC addressing and switching; Layer 3 (Network): IP addressing and routing; Layer 7 (Application) : AV software and protocols	The OSI model provides a framework for diagnosing where in the network stack an AV problem occurs. A device that shows connected (Layer 1) but cannot communicate (Layer 3) has a different problem than one with no link light (Layer 1). AV professionals need to converse across layers to collaborate effectively with IT staff.	Domain 5	2.5 Designer
2	A — IPv4 addresses consist of four octets (e.g., 192.168.1.1) ; the subnet mask defines which portion is the network address and which is the host address	AV devices must be assigned IP addresses within the correct subnet to communicate. A device assigned 192.168.1.50/24 cannot communicate with a device at 192.168.2.50/24 without routing, even	Domain 5	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		if they are physically connected. Understanding subnets prevents one of the most common AV-over-IP installation errors.		
3	B — DHCP assigns IP addresses dynamically from a server pool; static assignment fixes a device's IP address permanently in its configuration	AV devices benefit from static IP assignment because their addresses must be known to control systems, monitoring software, and network documentation. DHCP is appropriate for user devices that move between networks. A hybrid approach (DHCP with reservations by MAC address) achieves both manageability and predictability.	Domain 5	2.5 Designer
4	D — VLANs (Virtual Local Area Networks) create logically separate	AV-over-IP systems, particularly audio transport protocols like Dante,	Domain 5	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	network segments on shared physical infrastructure, allowing AV traffic to be isolated from general network traffic	generate multicast traffic that can saturate flat networks. Placing AV devices on a dedicated VLAN prevents this traffic from impacting general network users while maintaining the shared physical infrastructure. VLAN design should be coordinated between AV and IT staff.		
5	A — QoS (Quality of Service) prioritizes time-sensitive AV traffic (audio, video) over less time-sensitive data (file transfers, web browsing) to prevent dropouts caused by network congestion	Without QoS configuration, audio and video packets compete equally with all other network traffic. Under load, packet delay and loss cause audible dropouts or visible frame drops in AV-over-IP systems. QoS policies ensure that AV traffic is processed preferentially at network switches, maintaining stream quality	Domain 5	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		during periods of heavy general network use.		
6	C — Dante is an AV-over-IP protocol developed by Audinate that transports multichannel professional audio over standard Ethernet networks	Dante is the most widely adopted professional audio networking protocol in educational AV, supported by hundreds of hardware devices from multiple manufacturers . It operates over standard network infrastructure but requires specific VLAN and QoS configuration to function reliably at scale. Dante is distinct from NDI (which transports video) and from AES67 (which is the open standard for audio-over-IP interoperability).	Domain 5	2.5 Designer
7	B — NDI (Network Device Interface), developed by NewTek/Vizrt , is a	NDI enables camera outputs, capture cards, and software applications to share video streams over a	Domain 5	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	protocol for transporting video over standard IP networks with low latency	LAN without dedicated video distribution infrastructure. It is widely used in modern hybrid classrooms for flexible camera routing. Unlike professional SDI infrastructure, NDI leverages existing network investments but requires adequate network bandwidth (typically 100–200 Mbps per stream for full-quality NDI).		
8	A — Default credential management : change manufacturer default usernames and passwords immediately on installation; network segmentation: place AV devices on isolated VLANs with firewall rules; firmware updates:	Network-connected AV devices represent a significant and often overlooked cybersecurity risk in educational institutions. Default credentials on AV devices are publicly documented by manufacturers and actively scanned for by malicious actors. A	Domain 5	2.3 Citizen

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	apply tested updates to patch security vulnerabilities	disciplined security posture for AV devices mirrors the baseline IT security practices applied to servers and workstations.		
9	C — Wi-Fi is appropriate for control signals and BYOD content sharing; high-bandwidth uncompressed AV-over-IP (Dante, NDI) requires wired Ethernet for reliability	802.11 Wi-Fi uses contention-based access that introduces variable latency — acceptable for web browsing but problematic for latency-sensitive professional audio. Wired Ethernet provides deterministic latency essential for professional AV performance. This distinction is commonly misunderstood by IT departments accustomed to treating all network traffic equally.	Domain 5	2.5 Designer
10	D — Effective AV-IT collaboration	The AV-IT convergence has made	Domain 5	2.4 Collaborator

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	requires shared understanding of each other's requirements, joint planning for infrastructure decisions, and clear governance defining responsibility boundaries	traditional departmental silos increasingly problematic. AV systems depend on IT infrastructure (networks, security policies, DHCP servers) while IT departments are unfamiliar with AV performance requirements. Formal collaboration protocols, joint planning meetings, and shared documentation systems are organizational solutions that prevent technical problems from becoming support failures.		

Chapter 6 Answer Key: Control Systems

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	B — Control system components: processor (the "brain"), user	Understanding the component architecture of a control system is	Domain 6	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	interface (touch panel, button panel, app), control interfaces (RS-232, IR, relay, IP), and feedback devices (status indicators)	prerequisite to both specifying and troubleshooting it. The processor executes logic; the user interface receives user input; control interfaces send commands to controlled devices; feedback devices confirm device status. Each component can be a point of failure independently of the others.		
2	A — RS-232 (serial, point-to-point device control), RS-485 (serial, multi-drop), TCP/IP (network-based control), relay (dry contact switching), IR (infrared command emulation)	Control system protocol selection depends on the controlled device's available interfaces. Most professional AV equipment supports RS-232, Ethernet, or both. Consumer devices may only support IR. Understanding which protocols each device supports is	Domain 6	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		essential for system specification and is documented in each device's control driver/module library.		
3	C — A macro executes a predefined sequence of commands from a single trigger event (button press, scheduled time, sensor input)	Macros are the primary programming tool in AV control systems, allowing complex multi-device sequences to be executed with a single user action. A "Start Class" macro might power on the display, set the input to laptop, unmute audio, set volume to 40%, and dim room lights — replacing six separate manual actions with one button press.	Domain 6	2.5 Designer
4	B — Centralized: one processor controls all devices in a room or zone; distributed: multiple	Centralized architectures are simpler to configure and maintain for single-room applications; distributed architectures provide	Domain 6	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	processors share control responsibilities; distributed systems are more resilient to single-point failure	redundancy and scalability for campus-wide systems. The failure mode differs critically: a failed central processor takes the entire system offline; a failed node in a distributed system affects only its assigned devices.		
5	D — Touch panel UX should prioritize simplicity (only necessary functions visible), clear labeling (no cryptic abbreviations), appropriate button sizing, and meaningful feedback (confirmation that a command was received)	The most technically capable control system fails its purpose if instructors avoid using it. User experience research in AV consistently identifies overly complex touch panels as the primary cause of AV non-use. Fewer, clearly labeled buttons with logical grouping outperform dense, comprehensive panels in real-world educational deployment.	Domain 6	2.5, 2.6

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
6	A — Feedback devices include LED indicators, on-screen status messages, and audio confirmation tones that confirm to the user that a command was received and executed	Feedback is a fundamental UX principle: users who press a button and receive no confirmation will press it again (and again), often causing unintended system states. Well-designed control systems provide immediate visual or auditory feedback for every user action. Lack of feedback is both a UX failure and a frequent cause of support calls.	Domain 6	2.5 Designer
7	C — Cloud-connected control systems enable remote monitoring, configuration, and maintenance but introduce network dependency; a cloud service outage can disable local control functionality	Cloud control offers genuine value for remote management and proactive monitoring but requires careful architectural consideration of failure modes. A best practice is to design systems that function locally when cloud connectivity is	Domain 6	2.5, 2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	if not designed with local fallback	unavailable — local control logic should not depend on cloud services for basic room operation.		
8	B — Control system troubleshooting follows the same eliminate-isolate-replace methodology: verify the user interface is sending commands; verify the processor is receiving them; verify the controlled device is responding; verify the device's physical state	A non-functioning macro can fail at any stage: the touch panel may not be sending the command; the processor may be receiving but not executing it due to a logic error; the controlled device may be receiving commands but failing to execute them due to a different input being selected or a hardware fault.	Domain 6, 10	2.7 Analyst
9	A — System presets (standardized configurations saved as macros) reduce operator error, ensure consistent system states, and make it easier for	Standardized presets are particularly valuable in educational environments where diverse end users — including substitute teachers and students — must operate AV systems. A "Laptop	Domain 6	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	substitute teachers or unfamiliar users to operate AV systems successfully	Presentation" preset that reliably delivers the correct display, audio, and lighting configuration reduces support calls and improves instructional continuity.		
10	D — RS-232 remains in active use in professional AV because it provides deterministic , low-latency, point-to-point device control without the complexity of network-based protocols	Despite its age, RS-232 serial control is still specified on current professional AV equipment because it is reliable, well-understood, and does not depend on network infrastructure. It is particularly valued in control applications where consistent, low-latency command execution is critical (projector control, display switching). Network-based control is more flexible but introduces network	Domain 6	2.5 Designer

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		dependency and potential latency variation.		

Chapter 7 Answer Key: Collaboration Technology

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1	B — Hardware room system	A dedicated room system integrates a purpose-built codec, echo-canceling microphone array, and camera optimized for a specific room — delivering more consistent performance than software-only solutions on general-purpose laptops.	Domain 7	2.4
2	C — Multiple cameras covering both instructor and student seating areas	Equity for remote participants requires that they can see both the instructor and student interactions. A single front-of-room camera excludes	Domain 7	2.3, 2.6

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		student-to-student discussion from the remote experience, creating an unequal learning environment.		
3	B — Wireless content sharing/presentation	Mersive Solstice is an enterprise wireless presentation platform enabling BYOD content sharing to room displays without physical connections. It competes with Barco ClickShare, Crestron AirMedia, and native wireless protocols (AirPlay, Miracast).	Domain 7	2.5
4	B — Using multiple capsules and signal processing to focus audio pickup toward active speakers	Beamforming arrays continuously steer their pickup pattern toward the detected speaker's location, providing clear audio capture across a room without requiring a dedicated microphone	Domain 7	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		for each participant. This is the standard microphone technology for modern collaboration room systems.		
5	C — Failure in complex scenes with multiple people or rapid movement	AI tracking algorithms perform well in typical lecture scenarios but fail when multiple people are equally active, when participants move rapidly or unpredictably, or when the algorithm's training data does not represent the room's population diversity. Physical camera placement remains the primary design decision.	Domain 7	2.7
6	B — Equivalent learning experience for in-room and remote participants	Hybrid equity addresses the documented disparity in educational experience between in-room and remote learners. A room system	Domain 7	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		that projects a remote participant's video as a small thumbnail in the corner creates fundamentally unequal participation — a design failure with pedagogical consequences.		
7	C — 1.5 × screen height	The 1.5H minimum distance in ANSI/AVIXA 2M prevents excessive neck strain and visible pixelation for viewers seated very close to large displays. Both the minimum and maximum viewing distances define the audience area that AV design must serve.	Domain 8	2.5
8	C — Design Development	Design development is the phase when schematic concepts are refined into specific equipment lists, signal flow diagrams, rack layouts,	Domain 8	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		and installation drawings. Commissioning is the post-installation verification phase; programming is the needs assessment phase; schematic design is the early concept phase.		
9	B — More even audio coverage across all seating positions	Distributed speaker systems with multiple smaller speakers mounted throughout the ceiling provide more uniform SPL across an audience area than a single point-source speaker. This reduces the SPL differential between front and rear seats, improving intelligibility equity across the room.	Domain 8	2.5
10	C — End users (teachers, students with disabilities) who will operate and learn in the	The programming phase should center the people who will live with the system daily. Facilities managers, IT	Domain 8	2.4

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	space	staff, and architects are necessary participants — but excluding end users from requirements gathering leads to systems designed against theoretical use cases rather than actual pedagogical needs.		

Chapter 8 Answer Key: Classroom AV Design

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1-2	1.5H minimum; 6H maximum per ANSI/AVIXA 2M:2010	ANSI/AVIXA 2M:2010 establishes viewing distance limits based on screen height and content type. These thresholds ensure that all seats in a properly designed room allow adequate image resolution and viewing comfort. Applying these standards	Domain 8	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		during the design phase prevents the common mistake of undersized displays in large rooms.		
3	Sight line calculation: room depth ÷ 6 = minimum screen height	For a 48-foot deep room, minimum screen height = $48 \div 6 = 8$ feet. This calculation is the foundational tool for display specification in any lecture-format room. Rooms with non-standard configurations (tiered seating, curved rows) require modified sight line analysis accounting for elevation and horizontal offset.	Domain 8	2.5
4-5	Dispersion angle matched to audience width; multiple speakers in deep rooms	Speaker coverage requires matching the horizontal dispersion angle (–6 dB beamwidth) to the audience width at the listening distance. Deep rooms may require delay	Domain 8	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		speakers to maintain intelligibility at the rear without excessive SPL at the front. Coverage uniformity (within ± 6 dB across the audience area) is the design target.		
6	Programming → Schematic Design → Design Development → Construction Documents → Commissioning	Each phase of the AV design process has defined deliverables and decision gates. Skipping phases (particularly programming/needs assessment) is the most common cause of AV systems that technically function but fail to meet educational needs. Commissioning — verifying performance against the design specification — is frequently omitted and leads to undetected deficiencies.	Domain 8	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
7	Performance , reliability, maintainability, total cost of ownership, and vendor support — not brand preference or initial price alone	Equipment selection criteria must be documented before evaluating specific products to prevent bias. Total cost of ownership (purchase + installation + maintenance + replacement) consistently provides a more accurate investment picture than initial cost alone. Maintainability — availability of parts, local service, and documentation — is particularly important in educational settings with limited AV staff.	Domain 8	2.5, 2.7
8	Rack design organizes AV equipment for accessibility, cooling, cable management , and serviceability	Rack layout should provide adequate spacing for heat dissipation, logical signal flow grouping, clear labeling, accessible cable management, and front/rear	Domain 8	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		access for service. Poorly designed racks are a major cause of overheating faults and extended service time when components require replacement.		
9	End users, disability services, facilities, IT, and administration — all should be consulted; end users are most frequently excluded	The absence of end-user voices in AV design is documented as the primary driver of systems that are technically functional but pedagogically inappropriate. Disability services consultation ensures accessibility requirements are designed in rather than retrofitted. A formal stakeholder consultation process should be part of every institutional AV project.	Domain 8	2.4
10	Energy-efficient displays, laser light sources (no lamp waste),	AV systems in large educational institutions represent significant	Domain 8	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	intelligent occupancy-based power management , and longevity-focused equipment selection reduce AV's environmental footprint	energy consumers and e-waste generators. Designing for longevity (selecting durable, maintainable equipment) and energy efficiency (displays with automatic brightness adjustment, occupancy-based power-off) aligns AV investment with institutional sustainability commitments.		

Chapter 9 Answer Key: Digital Signage

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1-2	Media player, CMS, display (network optional but typical)	The three core components of a digital signage system are the media player (hardware that stores and plays content), the CMS (software that manages, schedules, and distributes content), and the display.	Domain 9	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		Network connectivity enables remote content management. SoC (System-on-Chip) displays integrate player and display into one device.		
3	8-12 seconds per content zone per slide	Dwell time (the duration each content slide is displayed) must provide sufficient time for a viewer walking past the sign to read and understand the content. Content complexity, reading speed, and viewer proximity all affect optimal dwell time. Overly short dwell times cause content to be unread; overly long dwell times make playlists feel static.	Domain 9	2.5
4	Zones are screen regions; playlists are ordered content sequences;	CMS scheduling tools give operators fine-grained control over what is shown,	Domain 9	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	dayparting schedules different content for different times of day; conditional triggers display content based on data inputs	where, and when. A campus CMS might display morning announcements in Zone A, dining menus in Zone B during lunch hours, and emergency alert content across all zones when triggered. Understanding these scheduling primitives is prerequisite to effective CMS operation.		
5	WCAG 2.1 contrast minimums (4.5:1 for text), text alternatives for non-text content, no content that flashes more than 3 times per second (seizure risk)	Campus digital signage is a public-facing information system with accessibility obligations under ADA Title II. WCAG 2.1 was developed for web content but provides the closest applicable framework for dynamic display accessibility. Interactive kiosks face stricter requirements because they serve as an interface to	Domain 9, 11	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		services — keyboard or switch access is required.		
6	EAS integration pre-empts all scheduled content across the entire signage network with emergency messaging	Emergency alert system integration allows authorities to push emergency notifications to all networked displays simultaneously, overriding scheduled content. This is a safety-critical function that should be designed, tested, and documented separately from routine content management. Response time from alert trigger to display on all screens should be verified during commissioning .	Domain 9	2.5, 2.3
7	General rule: 1 inch of letter height per 10 feet of viewing distance for body text; approximately 30-point font for 10-	Font size for digital signage follows the same physical principles as display image size: the angular size of text must be sufficient for	Domain 9	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	foot viewing distance on a standard display	the viewer's acuity at the viewing distance. Body text on a sign viewed at 10 feet should be readable without close approach; headline text should be visible from further away to attract attention.		
8	Central IT/marketing approval for brand standards + departmental edit access within pre-approved templates	Effective digital signage governance balances institutional brand consistency with departmental information autonomy. A tiered permission model — where central administrators control templates and brand elements while departments control content within those templates — addresses both needs without requiring central review of all content changes.	Domain 9	2.4

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
9	ADA requires interactive kiosks to be operable without fine motor control; reach ranges must be within ADA limits; touchscreen-only kiosks without alternative input are non-compliant	Interactive digital kiosks in public educational spaces must comply with ADA Section 4.34 (transaction machines) and emerging guidance on self-service technology. The primary requirements are reachable controls (forward reach: max 48", side reach: max 54"), and operable controls that do not require tight grasping, pinching, or twisting. Touch-only interfaces without alternative input methods are non-compliant.	Domain 9, 11	2.3
10	Location (high-traffic, eye-level placement), relevance (content matching the audience and location), simplicity (limited text, high	Research on campus wayfinding and digital signage effectiveness identifies placement and content relevance as the strongest predictors of sign	Domain 9	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	contrast), and freshness (regularly updated content)	noticeability and message retention. Signs in low-traffic locations, displaying irrelevant institutional content with poor contrast, are functionally invisible to their intended audience regardless of display quality.		

Chapter 10 Answer Key: Maintenance and Troubleshooting

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1-2	Define the problem → Gather information → Identify possible causes → Test (eliminate) → Implement solution → Verify → Document	The 7-step methodology ensures systematic rather than intuitive troubleshooting. Each step has a specific purpose: definition prevents solving the wrong problem; information gathering prevents premature	Domain 10	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		conclusions; elimination testing avoids unnecessary replacement; documentation prevents recurrence of the same fault without institutional memory.		
3	Power (no output); signal (wrong input or broken connection); control (system not responding); display (hardware fault); audio (muted, wrong source, failed amplifier); network (device offline)	Categorizing faults before beginning diagnosis focuses the investigation. A power fault manifests differently from a signal fault: a display with no image but a functioning power LED has a signal issue, not a power issue. Systematic categorization prevents technicians from replacing expensive components that are not the source of the fault.	Domain 10	2.7
4	Cleaning (filters, vents, lenses), firmware updates, lamp/lamp hour	Preventive maintenance categories map to the most common failure modes: projector lamp failure	Domain 10	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	monitoring, battery replacement, cable inspection, display calibration, control system testing	(preventable with hour tracking), display degradation (preventable with calibration), connectivity failures (preventable with cable inspection), and cybersecurity vulnerabilities (preventable with firmware updates). A documented PM schedule converts these preventions into systematic practice.		
5	Track actual lamp hours in the projector's maintenance menu; replace at 80% of rated life to prevent unexpected mid-class failure	Projector lamp failure is the most common preventable AV fault in educational settings. Replacing lamps at 80% of rated hours rather than waiting for failure eliminates mid-class outages. Lamp hour data should be part of the regular PM record and reviewed at each inspection.	Domain 10	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
6	As-built signal flow diagrams, equipment schedules (make/model/serial/location), warranty records, and work order history	The four core documentation types form the institutional memory of an AV system. As-built drawings reflect what was actually installed (which may differ from design drawings); equipment schedules enable rapid identification of warranty status and spare parts needs; work order history reveals recurring faults that may indicate systemic design problems.	Domain 10	2.7
7	Escalate when the fault requires work inside energized equipment, structural modification, specialized test equipment, or expertise beyond the staff member's training and authority	Clear escalation criteria protect both the AV system and the staff member. Non-certified personnel working inside energized equipment or on structural mounts create safety and liability risks. A formal escalation	Domain 10	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		protocol (documented in a support response plan) ensures faults are routed appropriately without delay.		
8	Read release notes before updating; test in a non-production environment first; document the current version before updating; have a rollback plan if the update introduces issues	Firmware updates can change device behavior in ways that break existing control system programming or introduce new incompatibilities. Educational AV systems are not appropriate for untested firmware deployment — a malfunctioning control system during class disrupts learning for the duration of the fault.	Domain 10	2.7
9	Preventive maintenance has higher fixed costs but lower total cost of ownership; reactive maintenance has lower short-term cost but higher total	The economic case for preventive maintenance in a 50-room school district is compelling: projector lamp failure costs approximately \$200–400 in parts plus labor and	Domain 10	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	cost including emergency repair premiums, extended downtime, and accelerated equipment replacement	scheduling disruption. A PM program that prevents all lamp failures in 50 rooms annually saves significantly more than its implementation cost. Downtime cost — lost instructional time — multiplies the economic impact further.		
10	Define roles (who is first responder, escalation contact, and vendor liaison), response time expectations, information to gather before calling for escalation, and documentation requirements	An effective AV support response protocol converts institutional knowledge into a reproducible process. New staff can follow the protocol without institutional memory. Published response time standards set accountability expectations. Pre-escalation information gathering (fault description, equipment serial numbers, steps already	Domain 10	2.4

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		taken) makes every escalation more efficient.		

Chapter 11 Answer Key: Accessibility in AV

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1-2	ADA Title II: public educational institutions; Section 508: federally funded entities; both require proactive, not merely reactive, accessibility	ADA Title II applies to all state and local government entities, including public K-12 schools and community colleges. Section 508 applies to federal agencies and their contractors. Both legal frameworks require that programs and services — including AV-mediated instruction — be accessible to people with disabilities, not merely available in an alternative format on request.	Domain 11	2.3
3	Hearing loop (induction loop)	Induction loop technology is the only	Domain 11	2.3, 2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	generates an electromagnetic field that is received by hearing aids and cochlear implants with T-coil (telecoil) mode enabled	assistive listening system that does not require a separate receiver device — hearing aid users with T-coil simply switch modes. IEC 60118-4 specifies the magnetic field strength requirements for compliant loop installations. ADA requires assistive listening systems in assembly areas seating 50 or more, with a specified minimum number of receivers for other technologies.		
4	Pre-recorded video content: captions required immediately (ADA, Section 508); live/real-time content: CART or ASR captioning required for accessibility; auto-	The legal distinction between pre-recorded and live captioning reflects technical feasibility: pre-recorded content can be captioned before distribution with 100% accuracy; live	Domain 11	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	generated captions alone may not satisfy accessibility requirements if accuracy is insufficient	captioning requires real-time technology (CART stenography or ASR). Accuracy matters: WCAG 2.1 Success Criterion 1.2.2 requires captions for pre-recorded audio in synchronized media; auto-captions with high error rates may not satisfy this criterion.		
5	WCAG 2.1 SC 1.4.3 (contrast): 4.5:1 for normal text; SC 1.4.1 (use of color): information not conveyed by color alone; SC 2.3.1: no content flashing more than 3 times per second	Applying WCAG 2.1 to AV interfaces (touch panels, digital kiosks, software UIs) ensures that users with visual impairments, color blindness, and photosensitive conditions can access control and information interfaces. Not all WCAG criteria apply directly to non-web AV interfaces, but contrast, seizure risk, and color	Domain 11	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		independence criteria are universally applicable to any visual display.		
6	UDL provides multiple means of representation (audio + visual + text), multiple means of engagement (flexible participation modes), and multiple means of action/expression (varied response formats) — principles that map directly to accessible AV design	Universal Design for Learning, developed at CAST, aligns with the AV designer's goal of creating systems that serve the widest range of users. An AV system that provides audio, visual, and captioned content simultaneously — without requiring users to request accommodation — embodies UDL principles. This reframes accessibility as proactive design rather than reactive compliance.	Domain 11	2.3, 2.5
7	Audio description is a narration of visual information (on-screen action, scene changes, graphics) for users who	Audio description is required for pre-recorded video content with visual information that is not conveyed through dialogue	Domain 11	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	are Blind or have low vision, provided during natural pauses in dialogue	(WCAG 2.1 SC 1.2.5, Level AA). It differs from captions (which serve users who are Deaf) and from audio-only content (which removes visual elements entirely). Produced audio description tracks should be available by default or selectable in accessible media players.		
8	Accessibility audit: systematic evaluation of a space against applicable standards (ADA, WCAG, ANSI A117.1) to identify gaps and prioritize remediation	The structured accessibility audit methodology provides defensible, documented evidence of current compliance status and a prioritized remediation roadmap. Audits should be conducted at initial installation (commissioning), when spaces are renovated, and periodically (recommended every 3-5	Domain 11	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		years) to account for equipment degradation and standard updates.		
9	The social model locates disability in the environment (barriers in design) rather than the individual; applied to AV, it means that a system that is inaccessible has a design defect, not a user defect	The social model of disability contrasts with the medical model, which locates disability within the individual as a condition to be treated. In AV design, the social model means that when a user who is Deaf cannot access audio content because there are no captions, the AV system — not the user — has failed. This reframing drives proactive accessible design rather than reactive accommodation.	Domain 11	2.3
10	Prioritize by: life safety (emergency notifications), regulatory compliance (legal obligations), high-impact-	Accessibility remediation is rarely a single-project activity — it requires prioritization against limited budgets. Life safety	Domain 11	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	on-learning deficiencies, and cost-effectiveness (highest impact per dollar)	deficiencies (inaccessible emergency notification systems) should always be addressed first regardless of cost. Legal compliance deficiencies follow. Educational impact and cost-effectiveness guide subsequent prioritization. A documented remediation plan with timelines demonstrates good-faith compliance effort.		

Chapter 12 Answer Key: Emerging Technologies

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
1-2	AI in AV: intelligent camera tracking, AI noise cancellation, ASR captioning, automated lecture capture,	AI applications in AV are evolving from assistive tools (noise cancellation, auto-framing) to autonomous systems (automated	Domain 1, 7	2.1, 2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	content summarization	lecture capture with speaker identification, AI-generated content summaries). Educational AV professionals must evaluate each application against evidence of learning outcome improvement, equity implications, and data privacy requirements — not merely technological novelty.		
3	VR (virtual reality): fully immersive; AR (augmented reality): overlays digital content on the physical world; MR (mixed reality): integrates physical and digital with real-world anchoring	Each XR modality has distinct educational applications: VR enables immersive simulations (surgery training, historical environments) ; AR enables contextual information overlays (anatomy labels on physical models); MR enables collaborative physical-	Domain 1	2.1, 2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		digital workspaces. Each also has distinct hardware requirements, content development costs, and accessibility implications.		
4	IoT sensors (occupancy, temperature, CO2, noise) enable automated AV management and provide data for informed room usage and maintenance decisions	IoT integration in educational AV extends from simple occupancy-based power management (turning off displays when rooms are empty) to sophisticated environmental monitoring (adjusting microphone sensitivity based on measured ambient noise) to predictive maintenance (tracking device usage hours automatically). Data governance and student privacy must be considered when IoT systems capture behavioral data.	Domain 1, 5	2.5
5	LED video	Micro-LED	Domain 3	2.5

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	walls use self-emissive LEDs; micro-LED uses LEDs at microscopic scale enabling very fine pixel pitch, higher brightness, and longer lifespan than OLED	technology is emerging as the next-generation display technology for premium educational venues, offering the black levels of OLED without burn-in risk, and luminance exceeding LCD. Current limitations include very high cost and manufacturing complexity at small pixel pitches. Commercial availability at educational price points is expected to increase through the late 2020s.		
6	Technology evaluation criteria: problem solved, for whom, cost and TCO, integration complexity, equity implications, evidence base, and sustainability impact	A rigorous technology evaluation framework prevents decisions driven by vendor marketing, peer adoption pressure, or novelty bias. The equity criterion — who will be able to use the technology independently	Domain 1	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		and who will not — is often absent from educational technology evaluations and is the contribution of this course's equity-centered philosophy to standard practice.		
7	AI systems trained on non-representative data perform less accurately for underrepresented groups; ASR accuracy is lower for accented speech; camera tracking may perform poorly for people with darker skin tones	AI bias in AV systems creates measurable equity gaps: if an AI microphone suppresses certain voice types as "noise," those voices are silenced in the hybrid learning environment. If a tracking camera loses a student with a darker skin tone more frequently, that student's participation is captured less reliably. These are not theoretical concerns — they are documented performance disparities requiring active	Domain 1, 11	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
		evaluation.		
8	LED video walls consume significant energy; micro-LED manufacturing produces electronic waste; newer is not automatically more sustainable — lifecycle analysis required	The sustainability profile of AV technology requires full lifecycle analysis: manufacturing (embodied carbon and resource extraction), operational energy consumption, and end-of-life disposal. A laser projector replacing 3 lamp projectors may reduce operational energy and lamp waste despite having a higher manufacturing footprint. Institutional sustainability commitments require AV professionals to engage with these tradeoffs.	Domain 1	2.3
9	Data generated by AV systems (room occupancy, participant identification, behavioral tracking) should be	AV systems with AI features increasingly generate personal data: camera systems can identify individuals,	Domain 1, 5	2.3

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	subject to FERPA, institutional privacy policies, and explicit data minimization principles	microphone systems can transcribe conversations, and occupancy sensors can track movement patterns. In educational settings, FERPA protections apply to student data. Institutional privacy policies should explicitly address AV-generated data collection, retention, and sharing before these systems are deployed.		
10	Future-proofing: design for flexible signal pathways, high-bandwidth structured cabling (minimum Cat6A), upgradeable processing, and AV-over-IP infrastructure that decouples hardware from	AV infrastructure designed around flexible principles — high-bandwidth cabling, IP-based distribution, and modular processing — accommodate s technology evolution without requiring complete infrastructure replacement. Proprietary	Domain 8	2.7

Q#	Answer	Explanation	AVIXA Domain	ISTE Standard
	topology	point-to-point installations often require costly full replacement when technology changes; standards-based AV-over-IP infrastructure enables component-level upgrades as technology evolves.		

Supplemental Resources

Guest Speaker Topics and Sourcing

Recommended Guest Speaker Topics by Chapter

Chapter	Guest Speaker Topic	Recommended Speaker Type	How to Source
Ch. 1 — AV Ecosystem	Career pathways in AV: from classroom coordinator to CTS-D designer	AVIXA member with CTS credential working in educational AV	AVIXA member directory at avixa.org ; local chapter contacts
Ch. 2 — Audio	Room acoustics in school design: what architects and AV engineers need to agree on	Acoustical engineer or architect specializing in educational facilities	National Council of Acoustical Consultants (NCAC); ASA member directory
Ch. 3 — Video	Display technology	AV systems	AVIXA member

Chapter	Guest Speaker Topic	Recommended Speaker Type	How to Source
	selection for large campus deployments	engineer from a regional integration firm	integrator directory; local NSCA chapter
Ch. 5 — Networking	AV-IT collaboration: how our university redesigned its network governance	University IT director or CTO with AV convergence experience	EDUCAUSE member institutions; local ITSC conference contacts
Ch. 7 — Collaboration	Designing hybrid classrooms for equity: lessons from post-pandemic renovation	Higher education learning space designer or facilities director	Society for College and University Planning (SCUP) members; ISTE network
Ch. 8 — Classroom Design	AV design from the user perspective: what teachers actually need	Experienced K-12 instructional technology coordinator	State instructional technology associations; ISTE member community
Ch. 11 — Accessibility	Nothing about us without us: disability advocates and AV system design	Disability rights advocate or accessibility consultant	National Association of the Deaf (NAD); AFB (American Foundation for the Blind); AHEAD
Ch. 12 — Emerging Tech	What's next: AI, XR, and the future of educational AV	AV industry futurist, researcher, or technology journalist	AVIXA Futures Council; InfoComm show session speakers; EdSurge contributors

Instructor Tip — Sourcing AVIXA Guest Speakers

Contact your local AVIXA chapter (chapters.avixa.org) at least 6 weeks before the planned session. AVIXA chapters frequently have members who volunteer for educational outreach. Brief all speakers on the course's vendor-agnostic policy before they agree to participate. A 5-minute pre-visit call to review expectations is

standard professional practice and prevents uncomfortable situations in class.

Virtual Guest Speaker Protocols

Many AV professionals are willing to participate virtually when in-person travel is impractical. For virtual guest sessions:

- Use a platform that supports screen sharing, as speakers frequently present diagrams or system examples
- Record the session (with speaker permission and student notification) for students who cannot attend synchronously
- Prepare 3–5 discussion questions in advance and share them with the speaker to enable more substantive responses
- Allow 10 minutes of unstructured student questions — these are often the most valuable part of the session
- Follow up with a written thank-you within 48 hours; maintain a speaker contact list for future cohorts

Field Trip and Virtual Visit Recommendations

Visit Type	Location/Resource	Best Course Timing	Learning Focus
In-Person: AV Integration Firm	A local AVIXA-member integration company's facility or active installation site (with permission)	Weeks 5–7	Signal flow, equipment racks, professional installation practices
In-Person: Campus Renovation Site	A classroom or lecture hall mid-renovation at your institution or a partner institution	Week 10	Classroom design, cable infrastructure, behind-the-wall AV installation
In-Person:	Local TV studio,	Weeks 3–4	Professional

Visit Type	Location/ Resource	Best Course Timing	Learning Focus
Broadcast/Production Studio	university media center, or corporate video production facility		audio/video signal chains, camera systems, live production
Virtual: AVIXA Xchange	avixa.org/xchange — AVIXA's online content and education platform	Ongoing	Industry news, on-demand education, career resources
Virtual: InfoComm Conference Session Recordings	InfoComm (annual AVIXA trade show) session recordings available via AVIXA Learning Portal	Weeks 9–12	Industry trends, emerging technology, case studies
Virtual: ISTE Conference Session Library	Conference recordings available via ISTE membership; free preview sessions available	Weeks 1–2, 11	Educational technology integration, equity in edtech, accessibility
Virtual: Manufacturer AV Showroom Tours	Many AV manufacturers offer virtual showroom tours showing product demonstrations in educational configurations	Any week	System demonstrations, product comparison (use with vendor-agnostic framing)

Virtual Visit Framing Note

When using manufacturer virtual showrooms or sponsored content, explicitly apply the course's vendor-agnostic evaluation framework. Assign students to identify what performance criteria the demonstrated system addresses and what questions they would need answered before specifying it. Manufacturer content is a valid source of product knowledge — it is not a substitute for needs-based evaluation.

Professional Development Resources for Instructors

AVIXA Professional Development

- **CTS Certification (avixa.org/cts):** Instructors who do not hold a CTS are strongly encouraged to pursue it. Teaching AV is significantly enriched by the credentialing process. AVIXA Premium membership provides access to the CTS Study Buddy AI learning tool.
- **AVIXA Learning Portal (avixa.org/training):** On-demand courses in audio, video, networking, and AV infrastructure. Modules align directly with course chapters and may be assigned as supplemental student reading.
- **InfoComm Conference (annual, June):** The world's largest professional AV tradeshow and conference. Instructors attending InfoComm earn AVIXA RUs for CTS renewal. Student registration is available at reduced rates.
- **AVIXA Diversity Council and TIDE Initiative:** Resources for instructors committed to equity, inclusion, and diversity in AV education. The TIDE (Technology, Inclusion, Diversity, and Equity) community hosts online and in-person events.

ISTE Professional Development

- **ISTE Conference and Expo (annual, June):** The premier event for educational technology. Particularly relevant for instructors teaching the equity, collaboration, and emerging technology chapters.
- **ISTE U (iste.org/isteu):** Online courses aligned to the ISTE Standards for Educators. Certificates available in digital learning design, computational thinking, and more.
- **ISTE EdTekHub (iste.org/explore):** Free practitioner-contributed resources, case studies, and standards implementation examples.

- **ISTE Standards Community (community.iste.org):** Discussion forums for educators implementing the ISTE Standards. A useful source of real-world examples for discussion questions.

Accessibility-Focused Professional Development

- **AHEAD (Association on Higher Education and Disability, ahead.org):** Annual conference and webinar series on disability services and accessible educational technology.
- **National Deaf Center (nationaldeafcenter.org):** Resources on Deaf and hard of hearing students in educational settings; directly applicable to the hearing loop and captioning content in Chapter 11.
- **Deque Systems Accessibility Training (deque.com/training):** WCAG 2.1 training applicable to digital signage and AV interface design covered in Chapters 9 and 11.
- **CAST UDL Guidelines (udlguidelines.cast.org):** Free interactive framework for applying Universal Design for Learning principles — directly relevant to Chapter 11 and the equity-centered course philosophy.

Connecting the Course to the AVIXA AV Technologist Certificate Pathway

This course is designed to prepare students for the AVIXA-Recognized AV Technologist Certificate as a first step toward the CTS credential. The alignment is as follows:

Course Component	AVIXA AV Technologist Preparation	Recommended Next Step
Chapters 1-6 content + Midterm	Covers core AV technology domains tested in the AV Technologist exam: audio, video, signal distribution, networking, control	Students scoring $\geq 85\%$ on Midterm should register for the AV Technologist Test via avixa.org
Lab reports and	Develops professional	Portfolio of lab reports

Course Component	AVIXA AV Technologist Preparation	Recommended Next Step
documentation	documentation skills valued in the CTS exam's application-level questions	serves as evidence of AV competency for AVIXA membership applications
Full 14-week course + Final Exam	Extends beyond AV Technologist scope into design, accessibility, and emerging technology — providing a stronger foundation for CTS preparation	Students completing the course with B or higher recommended to pursue CTS: AV Technology course at avixa.org/training
Capstone Project	Demonstrates applied AV design competency consistent with CTS-D (design) pathway knowledge areas	Outstanding capstone projects may be submitted as portfolio evidence to AVIXA for recognition; coordinate with ISTE/AVIXA liaison

Instructor Action — Credential Coordination

Contact your AVIXA regional representative (avixa.org/contact) to establish your program as an AVIXA Educational Partner. This status provides students with discounted AVIXA membership rates (which includes free access to the AV Technologist Test) and provides the program with access to AVIXA curriculum resources. The ISTE × AVIXA co-brand on this curriculum provides a strong foundation for that conversation.

Student Certification Pathway Timeline

Timeline	Recommended Action
Week 1	Students create a free AVIXA account at avixa.org and explore the Learning Portal and AV Technologist Test information
Week 6	Students who are interested in the AV

Timeline	Recommended Action
	Technologist Certificate join AVIXA as Premium members to access the free test and Learning Portal modules
Week 8 (after Midterm)	Students scoring $\geq 85\%$ on Midterm are encouraged to attempt the AV Technologist Test; instructor provides a practice session using sample questions
Week 14	Students completing the course with a grade of B or higher receive a letter of completion from the instructor suitable for AVIXA membership and credential applications
Post-course	Students pursuing CTS should enroll in AVIXA CTS 1: AV Technology course and complete the CTS Prep track (Audio 2, Video 2, Networking 2, Creating/Implementing/Servicing AV Solutions)

Recommended SME Reviewer Types

Future editions of this Instructor Guide and the co-branded textbook should engage Subject Matter Experts (SMEs) from the following categories to ensure accuracy, currency, and equity of content. Instructors using Edition 1.0 who identify content requiring updating or correction should submit feedback to the ISTE x AVIXA curriculum committee via the portal identified in their program agreement.

SME Category	Expertise Needed	Source Organizations	Review Priority
CTS-D (Certified AV Designer)	Technical accuracy of signal flow, system design, and equipment specification chapters (Ch. 4, 5, 8)	AVIXA member directory; InfoComm Education Advisory Council	High
Educational Acoustician	Chapter 2 acoustic content; ANSI/ASA S12.60 currency; classroom acoustic design standards	National Council of Acoustical Consultants; Acoustical Society of America	High

SME Category	Expertise Needed	Source Organizations	Review Priority
K-12 Technology Coordinator	Real-world applicability of labs and assessments; currency of classroom AV equipment examples	CoSN (Consortium for School Networking); ISTE member community; state ed-tech associations	High
Disability Rights/Accessibility Consultant	Chapter 11 accuracy; ADA and Section 508 currency; authenticity of disability-centered framing	AHEAD; National Federation of the Blind; NAD; AAPD	High
AV-over-IP / Networking Specialist	Chapter 5 networking content; currency of AV-over-IP protocols (Dante, NDI, AVB/TSN)	AVIXA Technology Standards Committee; SMPTE Education Committee	Medium
UC&C / Collaboration Technology Specialist	Chapter 7 currency; video conferencing platform evolution; hybrid learning research	UCToday industry analysts; BICSI members; Zoom/Teams/Cisco education teams	Medium
Digital Signage Practitioner	Chapter 9 CMS platform examples; signage accessibility guidance currency	Digital Signage Federation; AVIXA Digital Signage Committee	Medium
Emerging Technology Researcher	Chapter 12 currency; AI in AV accuracy; XR in education evidence base	EDUCAUSE Center for Analysis and Research; MIT Media Lab; Horizon Report authors	Medium
Community College Instructional Designer	Pedagogical validity of lesson plans and assessments; appropriateness for adult learners	League for Innovation in the Community College; NISOD members	Medium
ISTE Standards Expert	Accuracy of ISTE Standard	ISTE Standards team	Medium

SME Category	Expertise Needed	Source Organizations	Review Priority
	alignments throughout guide; currency of 2024 v4.02 framework	(standards@iste.org); ISTE Educator Standards Coaches Community	
Equity and Inclusion Specialist in EdTech	Equity framing throughout course; cultural responsiveness of examples; representation in case studies	ISTE Equity PLN; Digital Promise; National Equity Project	High
AV Maintenance Technician	Chapter 10 accuracy; realism of PM checklist; troubleshooting scenarios	NSCA (National Systems Contractors Association) technician membership; AV service organizations	Low

SME Diversity Note

The ISTE × AVIXA curriculum development team is committed to engaging SMEs who reflect the diversity of educators, students, and AV professionals this curriculum serves. Reviewers should include people with disabilities, members of historically underrepresented groups in STEM and AV professions, and educators from a variety of institutional contexts (rural, urban, large, small, well-resourced, under-resourced). A technically accurate curriculum that does not reflect its full audience is an incomplete curriculum.

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Feedback and Corrections	Submit content corrections and update suggestions to the ISTE × AVIXA curriculum committee via the instructor portal. Include the chapter number, section, page reference, and a brief description of the issue.

Item	Details
Next Scheduled Review	Edition 1.1 — May 2028 (biennial review cycle)



A Note from the Curriculum Team

Teaching a course that sits at the intersection of two professional fields — AV technology and educational practice — requires both technical knowledge and pedagogical craft. The instructors who use this guide are the bridge between the principles of the AV industry and the people who will learn, teach, and lead in technology-enabled educational spaces. Your work matters — not just to the students in your cohort, but to every learner who will eventually sit in a classroom, meeting room, or learning space designed by someone your students will become. Thank you for bringing this curriculum to life with the care, rigor, and equity it deserves.

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